

claude-windows / 54d4074a-5a8d-4058-b42d-a757c338132c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra\54d4074a-5a8d-4058-b42d-a757c338132c.jsonl`  
- SHA-256: `a2aab6971f8a38839cd74b07ae3a350a4839769158f80062de5e43f26a60177c`  
- Source modified: `2026-06-21T21:40:25+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `C--Users-avidu-Projects-khelsutra`  
- session_id: `54d4074a-5a8d-4058-b42d-a757c338132c`
```

Transcript

1. user (2026-06-21T19:01:19.426Z)

An adversarial translation audit of the shared i18n catalogs (repo-root `/locales/kn/common.json` and /locales/hi/common.json`) surfaced real quality bugs in the **SPA console** translations (the B5/B6 Kannada/Hindi work ? AI-drafted, native-review-pending). These are in the non-`site` namespaces (`app.*`, `bulk.*`, `grid.*`, `footer.*`, `result.*`, `ingest.*`, `errors.*`); the `site.*` marketing namespace is fine (A3). The en source of truth is /locales/en/common.json`. Scope this as a focused "native-review pass on the kn/hi SPA console catalog" PR.`

CRITICAL (correctness):

- Hindi transliterates the brand "KhelSutra" ? "?????????" in ``ingest.intro``, ``ingest.reassure``, ``ingest.fileHint`` (check ALL hi keys for this). Brand MUST stay Latin "KhelSutra".
- ``footer.floorSomeRest`` (both kn + hi): leaks an internal ``min_shot_count`` code token (kn even wraps it in a `<code>` tag) and the word "engine". The en is plain "(they're too short)". Fix kn ? "(??? ????? ??????????)". and hi ? "(?? ??? ???? ???)?" , dropping the token/tag.

JARGON LEAK (en deliberately avoids "engine"/"server"; both languages leak it):

- ``footer.floorOne``, ``footer.floorAll``: kn/hi say "engine (?????/????) will skip it"; en says "it would be skipped" (no engine). Rephrase passively without "engine".
- ``errors.networkError``: kn "????? ??????????????" / hi "???? ? ???? ??????" ? en is "Can't reach KhelSutra. Is it running?" Use the brand, not "engine".
- ``ingest.intro`` (kn + hi) and ``ingest.reassure`` (kn says "?????????????", hi "???? ?"): add bucket/engine/server clauses absent from en. Match en's plainer meaning.

HINDI FLUENCY (over-Sanskritized / stiff for non-tech coaches/parents):

- ``app.badge``, ``app.intro`` (awkward calque), ``grid.shots`/`grid.shotsTitle`` ("?????????"/"?????????"), ``grid.shotsConfidence`` ("?????" for confidence), ``query.support``, ``result.dismiss`` ("???? ??" ? "?? ??" / "????").

KANNADA:

- ``bulk.invert``: "?????????" means rotate, not invert-selection ? use "???? ??????".
- ``result.dismiss``: "?????????" is legalistic ? "?????".

Keep ALL `/ICU {count, plural, ?} / {min} / {count}`` placeholders and `<x>/<i>` tags intact. After editing, run ``cd web && npm ci && npm run typecheck && npm test`` (the hygiene key-parity + placeholder-parity + es 3-form tests must stay green) and the SPA i18n render tests. Branch from origin/master, one PR, merge on green CI. Tag the catalogs "AI-draft + adversarial-review pass; native review still pending."

2. assistant (2026-06-21T19:01:23.916Z)

I'll start by getting oriented ? reading the session handoff, pulling latest, and understanding the catalog structure before making any edits.

3. user (2026-06-21T19:01:24.863Z)

```

1      # Session handoff ? khelsutra (Friendly & Multilingual workstream)
2
3      _Last updated: 2026-06-21, after B3 merged._
4
5      ## You are here
6      - **B3 just merged** ([#70](https://github.com/avidullu/khelsutra/pull/70), squash
`1b1fa7b` on master): the marketing `site/` is now i18n-wired (shared `t()`/`data-i18n` shim +
persistent en/kn/hi switcher + `server.js` Accept-Language negotiation), and the catalogs are
**hoisted to repo-root `/locales`** (D-CATALOG-HOME). **en-only marketing content** so far. Web
173 + site 62 tests green; merged on green CI (web + site + Cloudflare).
7      - The SPA (`web/`) was already fully trilingual+ (en/kn/hi + es/da/id) before this.
8      - Built in an isolated worktree to avoid a concurrent **A9** session's catalog WIP ? see
[[shared-checkout-concurrency]].
9
10     ## Next steps / open threads (tracker §7 of docs/FRIENDLY_AND_MULTILINGUAL.md)
11     - **A3** ? translate the marketing copy (`site.*` namespace + recording-education) to
kn/hi. Currently en-seeded. Gated on D-FONT for site (kn/hi glyphs need Noto on the marketing
site too).
12     - **A2** WhatsApp share, **A6** plain-language Request-Access framing, **A7** recording
diagram.
13     - **Native-speaker review** of the AI-drafted kn/hi catalogs before a wide public
launch.
14     - Do NOT archive the project doc until BOTH surfaces render kn/hi *content* (not just
en).
15
16     ## Ramp-up kit (read these to get current)
17     - `docs/FRIENDLY_AND_MULTILINGUAL.md` ? §7 tracker is the source of truth; §4
architecture; §6.5 negotiation matrix (N1?N9); §8 decisions.
18     - `/locales/en/common.json` ? the shared catalog (SPA console keys + the `site.*`
marketing namespace).
19     - `web/src/i18n/index.ts` + `web/src/i18n/hygiene.test.ts` ? SPA wiring + CI guards.
20     - `site/public/i18n.js` (shim), `site/server.js` (negotiation), `site/scripts/copy-
locales.mjs` (the one build step), `site/src/{i18n.shim,locales-drift,assets,server}.test.js`.
21     - [[catalog-home-and-i18n-layout]] for the catalog layout + negotiation contract.
22
23     ## Key decisions (don't relitigate)
24     - **D-CATALOG-HOME**: catalogs at repo-root `/locales`; SPA imports them; site mirrors
via `copy:locales` + drift guard.
25     - Marketing content is **en-only first** (prove the architecture); kn/hi marketing = A3.
26     - Negotiation order is identical across surfaces: persisted (`khelsutra.lang`) ?
`?lang=` ? navigator/Accept-Language ? en; missing key ? en, never a raw key.
27     - Work in an **isolated git worktree** here (shared checkout, rapid master churn).
28

```

4. user (2026-06-21T19:01:25.323Z)

```

1      ---
2      name: shared-checkout-concurrency
3      description: khelsutra is a shared single checkout with rapid concurrent master churn;
isolate repo-writing work in a git worktree
4      metadata:
5        node_type: memory
6        type: project
7        originSessionId: dca1ff63-3aec-42ab-897f-41970003f4e1
8      ---
9
10     The `khelsutra` repo is worked by **multiple concurrent sessions on ONE shared working
checkout** (`C:/Users/avidu/Projects/khelsutra`). During the 2026-06-21 B3 session, master
advanced **5+ times** (#67?#68?#69?#71?#70) within ~1 hour, and a sibling **A9** session had
**uncommitted catalog WIP sitting in the shared working tree** that carried across a `git
checkout`.
11
12     **Why:** the owner runs several parallel "slates"; the shared tree + current branch can
shift under you between two commands, and sibling WIP must never be clobbered (a past incident
had a squash-merge silently absorb sibling work).

```

```

13
14     **How to apply (for any repo-writing task here):**
15     - **Don't work in the shared checkout when a sibling may be active.** Create an isolated
worktree: `git worktree add <path> -b <branch> origin/master` (or check out an existing branch).
Run `npm ci` per package there (web/ + site/ each have their own `node_modules`; not shared
across worktrees). Clean up after merge: `git worktree remove <path> --force`.
16     - **Always branch from `origin/master`** and re-verify `git log origin/master..HEAD`
shows ONLY your commits before opening a PR.
17     - **Stage explicit paths**, never `git add -A`.
18     - **Expect master to move mid-task.** If your PR shows `mergeable=false` but `git merge-
base --is-ancestor origin/master HEAD` is true, GitHub's mergeability is just **stale** ?
collapse to one clean commit on top of latest `origin/master` (`git reset --soft origin/master`
+ recommit) and `git push --force-with-lease` to force a fresh recompute (this also triggers the
`test` workflow, which GitHub **skips while a PR is CONFLICTING**).
19
20     See [[catalog-home-and-i18n-layout]] for the i18n-specific contention (catalog files).
21

```

5. user (2026-06-21T19:01:26.271Z)

```

1     ---
2     name: catalog-home-and-i18n-layout
3     description: "i18n catalogs live at repo-root /locales (D-CATALOG-HOME); SPA imports
them, site mirrors them via a copy step"
4     metadata:
5         node_type: memory
6         type: reference
7         originSessionId: dca1ff63-3aec-42ab-897f-41970003f4e1
8     ---
9
10     After B3 (#70, 2026-06-21), the localization catalogs are the **single source of truth
at repo-root `/locales/{en,kn,hi,es,da,id}/common.json`** (D-CATALOG-HOME). Both surfaces
consume them:
11
12     - **SPA (`web/`):** `web/src/i18n/index.ts` imports `../../../locales/<lng>/common.json`
directly. Vite resolves the workspace root via the repo `.git` marker, so dev/build/vitest all
read the parent-dir JSON. `web/src/i18n/hygiene.test.ts` reads `/locales` for key-parity +
placeholder-parity (+ an es 3-form plural check). Adding a key to en means adding it to ALL
locales or these tests fail.
13     - **Marketing site (`site/`, dependency-free):** `site/scripts/copy-locales.mjs` mirrors
`/locales` ? `site/public/locales/` (`npm run copy:locales`; `predeploy` hook). The committed
copy is what ships; `site/src/locales-drift.test.js` fails CI if it drifts. The shim
`site/public/i18n.js` (vanilla `t()/\`data-i18n`, ICU-lite via `Intl.PluralRules`) fetches those
at runtime; `server.js` does Accept-Language negotiation (cookie??lang?Accept-Language?en) and
stamps `

```

6. user (2026-06-21T19:01:31.049Z)

On branch master
Your branch is behind 'origin/master' by 1 commit, and can be fast-forwarded.
(use "git pull" to update your local branch)

Untracked files:

```
(use "git add <file>..." to include in what will be committed)
.claude/
site/access.db
```

nothing added to commit but untracked files present (use "git add" to track)

---BRANCH---

master

---LOG---

1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)

c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics) (#73)

1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-HOME) (#70)

7. assistant (2026-06-21T19:02:08.229Z)

Local master is 1 commit behind origin (A3 #74 just merged, touching the catalogs). Per the concurrency convention, I'll work in an isolated worktree branched from `origin/master`. Let me set that up and read the catalogs.

8. user (2026-06-21T19:02:09.915Z)

Preparing worktree (new branch 'fix/kn-hi-spa-console-native-review')

branch 'fix/kn-hi-spa-console-native-review' set up to track 'origin/master'.

HEAD is now at 1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)

---WORKTREE LIST---

C:/Users/avidu/Projects/khelsutra	c1848df [master]
C:/Users/avidu/Projects/khelsutra-fu-wt	1cd84af [feat/a6-access-plain-language]
C:/Users/avidu/Projects/khelsutra-i18n-review	1cd84af [fix/kn-hi-spa-console-native-review]

9. assistant (2026-06-21T19:02:13.498Z)

Worktree created (note: a sibling session is active on `feat/a6-access-plain-language`). Now let me read the three catalogs for the affected namespaces.

10. user (2026-06-21T19:02:14.082Z)

```
1      {
2        "app": {
3          "badge": "pick your rallies · alpha",
4          "title": "Pick the rallies for your reel",
5          "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-
6            tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
7        },
8        "lang": {
9          "label": "Language",
10         "en": "English",
11         "kn": "?????",
12         "hi": "?????",
13         "es": "Español",
14         "da": "Dansk",
15         "id": "Bahasa Indonesia"
16       },
17       "bulk": {
18         "selectAll": "Select all",
19         "clear": "Clear",
20         "invert": "Invert"
21       },
22       "mode": {
23         "demo": "These are sample rallies ? process a video above to work on a real match.",
24         "loading": "Finding the rallies in your video?",
```

```

24     "error": "Couldn't load this video's rallies: {error}",
25     "loaded": "Found <b>{count}</b> rallies in your video."
26 },
27 "query": {
28     "placeholder": "e.g. over 10s · top 15 longest · shortest first",
29     "ariaLabel": "Filter rallies by length, count or order",
30     "apply": "Apply",
31     "notAvailable": "not available yet",
32     "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
33     "notAvailableHelp": "Shot type, players and score aren't available yet ? filter by
rally length instead.",
34     "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
35     "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
36     "try": "Sample queries:",
37     "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
<x>experimental</x> (under development).",
38 },
39 "grid": {
40     "empty": "No rallies to show yet.",
41     "rally": "Rally {n}",
42     "inReel": "Rally {n}, in reel",
43     "notInReel": "Rally {n}, not in reel",
44     "start": "start",
45     "long": "long",
46     "shots": "~{count} shots · experimental",
47     "shotsTitle": "Shot count is experimental (under development) and may be
inaccurate{conf}. Filter by rally length instead.",
48     "shotsConfidence": " ? confidence: {conf}"
49 },
50 "footer": {
51     "count": "{count, plural, one {# rally} other {# rallies}}",
52     "summaryRest": "selected · {total} total",
53     "reelName": "Reel name",
54     "reelNamePlaceholder": "my-highlights",
55     "build": "Build reel ({count})",
56     "building": "Building?",
57     "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? it would
be skipped, so the build would fail. Pick a longer rally.",
58     "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ?
they'd all be skipped, so the build would fail. Pick longer rallies.",
59     "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
60     "floorSomeRest": "with fewer than {min} shots will be skipped when building (they're
too short).",
61 },
62 "result": {
63     "ready": "Your reel is ready",
64     "failed": "Build failed",
65     "building": "Stitching your reel?",
66     "buildingReassure": "Joining your selected rallies into one video ? this usually
only takes a moment.",
67     "dismiss": "Dismiss",
68     "download": "Download {filename}"
69 },
70 "build": {
71     "noVideo": "These are sample rallies ? process a real video first to build a reel."
72 },
73 "ingest": {
74     "title": "Process a video",
75     "intro": "Point KhelSutra at a video and it finds the rallies for you ? this can
take a few minutes.",
76     "uriPlaceholder": "gs://your-bucket/match.mp4",
77     "uriAriaLabel": "Video location ? a gs:// bucket URL or a path on the engine host",
78     "hint": "Paste a cloud-storage link (gs://?), or a path to a video on the machine
running KhelSutra.",
79     "submit": "Process video",

```

```

80     "submitting": "Submitting?",
81     "cancel": "Cancel",
82     "working": "Processing your video?",
83     "progress": "{progress}",
84     "retry": "Try again",
85     "reassure": "Safe to wait ? KhelSutra keeps working even if your connection is slow;
a long video can take several minutes.",
86     "statusFailed": "Couldn't process",
87     "chooseFile": "Choose a video file",
88     "uploadAria": "Choose a video file to upload",
89     "fileHint": "Pick an .mp4 or .mov from this device ? it uploads, then KhelSutra
finds the rallies.",
90     "orLink": "Or use a cloud link",
91     "uploadingStatus": "Uploading?",
92     "uploadProgress": "Upload progress"
93 },
94 "errors": {
95     "networkError": "Can't reach KhelSutra. Is it running?",
96     "invalidUri": "Enter a video location first ? a gs:// link or a path on the machine
running KhelSutra.",
97     "emptyResult": "Processing finished but found no rallies in this video. Try a
clearer match recording.",
98     "timeout": "This is taking longer than expected. The job may still be running ?
reload with the video link, or try again.",
99     "engineGeneric": "Something went wrong. Please try again."
100 },
101 "tap": {
102     "title": "Tap to filter:",
103     "longest": "Longest first",
104     "shortest": "Shortest first",
105     "over10": "Over 10 seconds",
106     "over20": "Over 20 seconds",
107     "top5": "Top 5 longest",
108     "top10": "Top 10 longest"
109 },
110 "site": {
111     "nav": {
112         "how": "How it works",
113         "why": "Why it's hard",
114         "ownModel": "Own your model",
115         "flyer": "Flyer",
116         "roadmap": "Roadmap",
117         "faq": "FAQ",
118         "requestAccess": "Request access"
119     },
120     "hero": {
121         "eyebrow": "Invite-only alpha · Bengaluru",
122         "lead": "KhelSutra finds every rally in your session and strips out the knock-ups,
walks and gaps ? so a full match becomes a clean highlight reel you can watch and download. No
editing, no scrubbing.",
123         "ctaPrimary": "Request alpha access",
124         "ctaHow": "See how it works"
125     },
126     "record": {
127         "kicker": "Record it right",
128         "title": "Five minutes of setup ? reliably great reels",
129         "intro": "Highlight quality starts with the recording. Set the camera up once like
this and KhelSutra has everything it needs to find the rallies.",
130         "goldenRuleLead": "The golden rule: one match, one court, in full frame ? and
don't move the camera.",
131         "goldenRuleRest": "A single court in frame gives the best results today; busy
multi-court footage is the hard case we're still sharpening.",
132         "place": {
133             "title": "Place the camera",
134             "body": "Behind <b>one baseline</b>, elevated (a high tripod or a balcony),

```

tilted down ~30° ? the whole court in frame, players never cropped."

```

135     },
136     "settings": {
137         "title": "Settings",
138         "body": "<b>1080p+</b>, landscape, <b>30 fps</b> for badminton (60 fps for table
139     },
140     "still": {
141         "title": "Keep it still",
142         "body": "Fixed tripod or mount. <b>No panning, no zooming</b> during play ? a
143     },
144     "upload": {
145         "title": "Record & upload",
146         "body": "Even lighting, then record the <b>whole session</b> and upload it as-is
147     },
148     "checklistCta": "Full 1-page capture checklist ?"
149 },
150 "capture": {
151     "title": "1-page capture checklist",
152     "intro": "Great highlights start with a steady, full-court recording. Five minutes
153     "ruleLead": "The golden rule: one match, one court, in full frame ? and don't move
154     "ruleRest": "Multi-court footage is the hard case we're still sharpening; a single
155     "step1": {
156         "title": "Place the camera",
157         "b1": "Behind <b>one baseline</b>, elevated ? a high tripod, a few metres up, or
158         "b2": "Tilt down about <b>30°</b>; keep the <b>whole court</b> in frame with a
159         "b3": "Players should never leave the frame."
160     },
161     "step2": {
162         "title": "Settings",
163         "b1": "<b>1080p or higher</b>, shot in <b>landscape</b>.",
164         "b2": "<b>30 fps</b> for badminton/pickleball; <b>60 fps</b> for table tennis.",
165         "b3": "<b>Audio ON</b> (don't mute) ? keep the mic unobstructed, wind/noise
166     },
167     "step3": {
168         "title": "Keep it still",
169         "b1": "Fixed tripod or mount.",
170         "b2": "<b>No panning, no zooming</b> during play."
171     },
172     "step4": {
173         "title": "Light & record",
174         "b1": "Even indoor lighting; avoid backlight, glare and very dark halls.",
175         "b2": "Record the <b>whole session</b> and upload it as-is ? KhelSutra does the
176     },
177     "doTitle": "Do",
178     "do1": "Fixed camera, full court, behind the baseline",
179     "do2": "1080p+, landscape, audio on",
180     "do3": "One match on one court",
181     "dontTitle": "Avoid",
182     "dont1": "Handheld or phone-in-hand",
183     "dont2": "Broadcast-style pans & zooming",
184     "dont3": "Several courts in one frame; very dark halls",
185     "consent": "Please record adults with their okay. Footage involving minors is
186     "questions": "Questions?",
187     "printTip": "Tip: press Ctrl/Cmd + P to print or save as PDF."

```

```
188     },
189     "ownModel": {
190         "requestAccess": "Request access ?"
191     }
192 },
193 "selection": {
194     "count": "{count, plural, one {# rally selected} other {# rallies selected}}"
195 }
196 }
197
```

11. assistant (2026-06-21T19:02:16.680Z)

Now the kn and hi catalogs (focusing on the SPA console namespaces):

12. user (2026-06-21T19:02:17.176Z)

```
1      {
2      "app": {
3          "badge": "?????-????? ???? . ????",
4          "title": "????? ?????????? ?????????? ??????",
5          "intro": "<b>????? ????</b>? ?????????? ?????? ???? ? ????. <i>???? ???? ????
15?</i> ? ??? ? ?????????????? ?????? ??? ? ?????????????? ?????????. ?????? <b>?????????</b>
?????????????????; ?????????????? ?????? ???????????"
6      },
7      "lang": {
8          "label": "????",
9          "en": "English",
10         "kn": "?????",
11         "hi": "?????",
12         "es": "Español",
13         "da": "Dansk",
14         "id": "Bahasa Indonesia"
15     },
16     "bulk": {
17         "selectAll": "????????? ??????????",
18         "clear": "?????????????",
19         "invert": "?????????"
20     },
21     "mode": {
22         "demo": "??? ????? ?????????? ? ?????? ?????????? ??? ???? ???? ???? ????
??????????????????",
23         "loading": "????? ?????????????? ?????????????? ??????????????????",
24         "error": "? ?????? ?????????????? ??? ??????????????: {error}",
25         "loaded": "????? ?????????????? <b>{count}</b> ?????????? ??????????"
26     },
27     "query": {
28         "placeholder": "???. over 10s · top 15 longest · shortest first",
29         "ariaLabel": "????, ????? ??? ???? ?????????? ?????????????? ?????? ????",
30         "apply": "?????????",
31         "notAvailable": "????? ??????????",
32         "notAvailableTitle": "?{token}? ?? {kind} ????? ??? ? ???? ??????????",
33         "notAvailableHelp": "???? ??????, ?????? ??? ???? ???? ?????????? ? ??????
????? ???? ?????????? ?????? ????",
34         "unreadable": "????? ?????????? ? ?over 10s? ??? ?top 15 longest? ??????????",
35         "hint": "10 ?????????????? ?????? ?????????????? over 10s ??? ???? ???? ? ????
????????? ?????? ????",
36         "try": "????? ??????????????",
37         "support": "<b>????? ???</b>? ?????????? ?????? ?????????? ??????????
????????????????? ????-????? <x>?????????</x> (?????????????????)."
38     },
39     "grid": {
40         "empty": "??????? ???? ?????? ??????????????",
41         "rally": "??????? {n}",
42         "inReel": "??????? {n}, ??????????",

```



```
43     "notInReel": "????? {n}, ??????????",
44     "start": "?????",
45     "long": "????",
46     "shots": "~{count} ?????? . ??????",
47     "shotsTitle": "???? ????? ????????????? (????????????) ?????
????????????{conf}. ?????? ?????? ?????? ?????? ?????? ????",
48     "shotsConfidence": " ? ??????: {conf}"
49 },
50     "footer": {
51         "count": "{count, plural, one {# ?????} other {# ?????????}}",
52         "summaryRest": "????????? . ????? {total}",
53         "reelName": "???? ?????",
54         "reelNamePlaceholder": "my-highlights",
55         "build": "???? ????? ({count})",
56         "building": "?????????????",
57         "floorOne": "<b>????? ??? ????</b> {min} ?????? ?????? ?????????
? ?????? ?????? ?????????????, ?????? ??? ??????????????. ?????? ????????????? ?????.",
58         "floorAll": "<b>????????? ????? {count} ??????????</b> {min} ?????? ?????
????????? ? ?????? ????????????? ?????????????, ?????? ??? ??????????????. ?????
????????????? ?????.",
59         "floorSomeLead": "{count, plural, one {# ?????} other {# ?????????}}",
60         "floorSomeRest": "{min} ?????? ?????? ????????????? ?????? ?????????????
(????????? <code>min_shot_count</code> ???)."
```

```
97     "emptyResult": "????????? ???????? ???? ? ???????????? ?????? ?????????? ??????????
????????? ?????? ?????????????? ?????????????.",
98     "timeout": "????????????? ?????? ??? ??????????????????????. ???? ?????? ??????????????????
? ?????? ?????????????? ???????? ???? , ???? ?????? ?????????????.",
99     "engineGeneric": "??? ??????????. ????????? ?????? ?????????????."
100 },
101     "tap": {
102         "title": "???????? ?????? ?????? ????:",
103         "longest": "??? ????????? ??????",
104         "shortest": "??? ????????? ??????",
105         "over10": "10 ?????????????? ??????",
106         "over20": "20 ?????????????? ??????",
107         "top5": "??? ?????? 5",
108         "top10": "??? ?????? 10"
109     },
110     "site": {
111         "nav": {
112             "how": "??? ???? ???? ??????????",
113             "why": "??? ??? ??????",
114             "ownModel": "???????? ??????",
115             "flyer": "?????????",
116             "roadmap": "?????????",
117             "faq": "?????????????",
118             "requestAccess": "???????? ????"
119         },
120         "hero": {
121             "eyebrow": "??????-????? ?????? . ??????????",
122             "lead": "KheISutra ?????? ?????????????? ????????????? ????????????? ??????, ????-????????,
????? ?????? ?????????????? ?????????????????? ? ?????? ?????? ????????? ?????? ?????????? ??????????
????? ?????? ???? ?????????? ?????? ?????????????? ??????.",
123             "ctaPrimary": "????? ?????? ??????",
124             "ctaHow": "??? ???? ???? ?????????? ??????"
125         },
126         "record": {
127             "kicker": "???????? ?????????? ????",
128             "title": "??? ?????? ???????? ? ?????????? ?????????? ??????????",
129             "intro": "??????? ???????? ?????????????????????? ?????????????????? ??????????????????
????? ?????????????? ???? ? ?????????????? ???????? KheISutra ?? ?????? ?????? ??????????",
130             "goldenRuleLead": "??????? ?????: ???? ?????, ???? ?????, ?????? ?????????????? ?
????? ?????????????? ??????????????",
131             "goldenRuleRest": "????? ?????? ?????????????? ?????? ???? ?????????? ????????
?????????; ???? ?????????? ?????? ?????? ???? ?????? ?????????????????? ???? ??????",
132             "place": {
133                 "title": "????????? ??????",
134                 "body": "<b>????? ??????????</b> ?????, ?????????? (????? ?????????? ???? ??????),
~30° ?????? ?????? ? <b>??? ??????</b> ?????????????, ???????? ?????? ??????????????????."
135             },
136             "settings": {
137                 "title": "?????????????????",
138                 "body": "<b>1080p+</b>, ?????????????????, ?????????????????? <b>30 fps</b> (?????
????????? 60 fps). <b>Audio ON</b> ? ?????? ?????? ???? ?????????????? ???????? ??????????????"
139             },
140             "still": {
141                 "title": "????????? ??????",
142                 "body": "????? ?????????? ???? ?????. ??? ?????????? <b>??????? ????, ???? ?????</b> ?
????????, ?????-????? ?????????? ? ?????????? ??????????????"
143             },
144             "upload": {
145                 "title": "????????? ???? ?????? ?????????? ????",
146                 "body": "????? ?????, ???? <b>??? ??????</b> ?????? ?????????? ???? ??????????
????????? ???? ? KheISutra ?????????????? ?????? ?????????? ???? ??????????"
147             },
148             "checklistCta": "????????? 1-????? ?????????? ?????????? ?????? ?"
149         },
150         "capture": {
```

```
151         "title": "1-???? ????????? ????????? ??????",
152         "intro": "????? ?????????? ?????????, ?????-????? ?????????????????? ???????????????.
??? ?????? ?????? ?????? ?????????????? ?????? ?????????? ??????????.",
153         "ruleLead": "?????? ???? : ??? ?????, ??? ?????, ????? ?????????????? ? ?????
????????????????? ??????????????.",
154         "ruleRest": "????? ?????????? ?????? ??? ?????? ?????????????????? ??? ?????; ???
?????? ?????????????????? ?????? ?????? ?????? ?????????? ??????????",
155         "step1": {
156             "title": "????????? ??????",
157             "b1": "<b>????? ??????????</b> ?????, ?????????? ? ?????? ?????????, ????? ?????
?????????????, ??? ?????????.",
158             "b2": "?????? <b>30°</b> ?????? ??????; ?????? ??? ?????? ??? ?????????
????????????? <b>??? ??????</b> ?????????????? ??????",
159             "b3": "????????? ?????? ?????????????? ??????????????."
160         },
161         "step2": {
162             "title": "?????????????????",
163             "b1": "<b>1080p ??? ??????</b>, <b>?????????????????</b>????? ??????????????",
164             "b2": "????????????????/????????????????? <b>30 fps</b>; ?????? ?????????? <b>60
fps</b>.",
165             "b3": "<b>Audio ON</b> (????? ??????) ? ??? ?????????????????? ?????, ???/????
????????? ??? ?????."
166         },
167         "step3": {
168             "title": "????????????? ??????",
169             "b1": "????? ?????????? ??? ??????",
170             "b2": "??? ?????????? <b>????????? ???, ??? ?????</b>."
171         },
172         "step4": {
173             "title": "????? ?????? ??????????",
174             "b1": "????? ?????? ?????; ?????????, ????????? ?????? ?????? ????????? ?????????????
?????????",
175             "b2": "<b>??? ?????</b> ?????? ?????????? ??? ?????????? ?????????? ??? ? ??????????
??????????? KhelSutra ??????????."
176         },
177         "doTitle": "?????",
178         "do1": "????? ?????????, ?????? ??????, ?????????? ??????",
179         "do2": "1080p+, ??????????????????, ?????? ???",
180         "do3": "????? ?????????????????? ??? ??????",
181         "dontTitle": "?????????",
182         "dont1": "????????? ?????? ??? ?????? ??????",
183         "dont2": "??????-????? ?????? ??????",
184         "dont3": "????? ?????????????? ??? ?????????????; ?????? ?????? ??????????",
185         "consent": "????????? ?????????? ??? ?????????????????? ?????????? ????. ?????????
????????????? ?????????????????? ?????????? ?????????? ?????????? ?????????? ?????????? ??????????.",
186         "questions": "?????????????????",
187         "printTip": "????? ?????????? ??? PDF ??? ?????? Ctrl/Cmd + P ??????"
188     },
189     "ownModel": {
190         "requestAccess": "?????? ??? ?"
191     }
192 },
193 "selection": {
194     "count": "{count, plural, one {# ?????? ?????????????} other {# ??????????
?????????????}}"
195 }
196 }
197
```

13. user (2026-06-21T19:02:20.927Z)

```
1     {
2         "app": {
3             "badge": "?????-???? ??? . ??????",
4             "title": "???? ??? ?? ??? ????????? ??????",
```

[illegible]

```
?? <code>min_shot_count</code> ????)?"
61     },
62     "result": {
63         "ready": "???? ??? ????? ??",
64         "failed": "????? ?????",
65         "building": "????? ??? ?????? ?? ??? ???",
66         "buildingReassure": "????? ?????? ??? ???????? ?? ?? ?????? ??? ?????? ?? ??? ?? ? ??????
?? ?? ??? ?? ?? ????? ?????",
67         "dismiss": "????? ?????",
68         "download": "{filename} ???????? ??????"
69     },
70     "build": {
71         "noVideo": "?? ?????? ???????? ??? ? ??? ?????? ?? ??? ?????? ?? ??? ?????? ????????
?????"
72     },
73     "ingest": {
74         "title": "??????? ???????? ?????",
75         "intro": "????????? ?? ?? ?????? ?????? ? ?? ?????? ?????? ?? ?????? ?? ?????? ?????? ? ?? ??
????????? ?????? ?????? ?????? ??? ?????? ?? ?????? ?????",
76         "uriPlaceholder": "gs://your-bucket/match.mp4",
77         "uriAriaLabel": "??????? ?? ?????? ? ?? gs:// ????? URL ?? ?????? ?????? ?? ?? ???",
78         "hint": "?? gs:// ????? ?????, ?? ?????? ??? ?????? ?? ???????? ?? ??? ?????? ??????",
79         "submit": "??????? ???????? ?????",
80         "submitting": "????? ?? ??? ???",
81         "cancel": "????? ?????",
82         "working": "????? ???????? ???????? ?? ??? ???",
83         "progress": "{progress}",
84         "retry": "??? ?? ?????? ?????",
85         "reassure": "????????? ?????? ?????????? ?? ? ?????? ???????? ?????? ?????? ?? ?? ?????????? ??????
?? ??? ?????? ?????? ??; ?????? ???????? ??? ??? ?????? ?? ?????? ?????",
86         "statusFailed": "????????? ?????? ?? ???",
87         "chooseFile": "??????? ?????? ?????",
88         "uploadAria": "?????? ?????? ?? ??? ???????? ?????? ??????",
89         "fileHint": "?? ?????? ?? ?? .mp4 ?? .mov ?????? ? ?? ?????? ?????, ??? ??????????
????????? ?????? ??????",
90         "orLink": "?? ???????? ?????? ?? ?????? ?????",
91         "uploadingStatus": "?????? ?? ??? ???",
92         "uploadProgress": "?????? ??????"
93     },
94     "errors": {
95         "networkError": "????? ?? ?????? ?????? ?? ?????? ?????? ?? ?? ??? ???",
96         "invalidUri": "????? ???????? ?? ?????? ?????? ? ?? gs:// ?????? ?? ?????? ?? ?? ?????",
97         "emptyResult": "????????????? ?????? ??? ?? ?? ???????? ?????? ?????? ?????? ?????? ?? ?????? ???
????????????? ??????????",
98         "timeout": "?? ?????????? ?? ?????? ??? ?? ??? ??? ??? ?????? ?? ?? ??? ?? ?????? ?? ? ??????
????? ?? ??? ?????? ?????, ?? ??? ?? ?????? ??????",
99         "engineGeneric": "??? ???????? ?? ??? ?????? ??? ?? ?????? ??????"
100     },
101     "tap": {
102         "title": "????????? ?????? ?? ??? ??? ?????:",
103         "longest": "????? ?????? ?????",
104         "shortest": "????? ?????? ?????",
105         "over10": "10 ?????? ?? ?????????",
106         "over20": "20 ?????? ?? ?????????",
107         "top5": "????? ?????? ??? 5",
108         "top10": "????? ?????? ??? 10"
109     },
110     "site": {
111         "nav": {
112             "how": "?? ?????? ??? ?????? ??",
113             "why": "?? ?????? ?????????? ??",
114             "ownModel": "????? ?????? ??????",
115             "flyer": "?????????",
116             "roadmap": "?????????",
117             "faq": "????????? ?????????",
```

[illegible]


```

Record<string, unknown>, key));
21     else keys.push(key);
22     }
23     return keys.sort();
24 }
25
26     const loadKeys = (locale: string): string[] =>
27         flatten(JSON.parse(readFileSync(join(LOCALES_DIR, locale, "common.json"), "utf8"))) as
Record<string, unknown>;
28
29     describe("i18n key-parity ? every locale has EXACTLY the en keys", () => {
30         const enKeys = loadKeys("en");
31         const others = readdirSync(LOCALES_DIR, { withFileTypes: true })
32             .filter((d) => d.isDirectory() && d.name !== "en")
33             .map((d) => d.name);
34
35         it("en is the non-empty source catalog", () => {
36             expect(enKeys.length).toBeGreaterThan(20);
37         });
38
39         for (const loc of others) {
40             it(`${loc} matches en exactly (no missing, no extra keys)`, () => {
41                 const keys = loadKeys(loc);
42                 const missing = enKeys.filter((k) => !keys.includes(k));
43                 const extra = keys.filter((k) => !enKeys.includes(k));
44                 expect({ locale: loc, missing, extra }).toEqual({ locale: loc, missing: [], extra:
[] });
45             });
46         }
47     });
48
49     // Key-parity proves every locale has the same KEYS; this proves every locale's VALUE
keeps the same
50     // simple `{placeholder}` set as en ? so a translation can never silently drop a
{count}/{error}/{min}/
51     // {filename}/{n}/? and break interpolation. (ICU plural headers like `{count, plural,
?}` carry no
52     // simple {word} placeholder and match as empty on both sides, so they're correctly
ignored.)
53     describe("placeholder parity ? every locale preserves en's {placeholders}", () => {
54         function flatEntries(obj: Record<string, unknown>, prefix = ""): Record<string,
string> {
55             const out: Record<string, string> = {};
56             for (const [k, v] of Object.entries(obj)) {
57                 const key = prefix ? `${prefix}.${k}` : k;
58                 if (v && typeof v === "object" && !Array.isArray(v)) Object.assign(out,
flatEntries(v as Record<string, unknown>, key));
59                 else out[key] = String(v);
60             }
61             return out;
62         }
63         const placeholders = (s: string) => (s.match(/\{(\w+)\}/g) ?? []).sort();
64         const load = (loc: string) => flatEntries(JSON.parse(readFileSync(join(LOCALES_DIR,
loc, "common.json"), "utf8")));
65         const enE = load("en");
66         const others = readdirSync(LOCALES_DIR, { withFileTypes: true })
67             .filter((d) => d.isDirectory() && d.name !== "en")
68             .map((d) => d.name);
69
70         for (const loc of others) {
71             it(`${loc} preserves every {placeholder} that en uses`, () => {
72                 const locE = load(loc);
73                 const mismatches: Array<{ key: string; en: string[]; got: string[] }> = [];
74                 for (const key of Object.keys(enE)) {
75                     if (!(key in locE)) continue; // missing keys are the key-parity test's job

```



```

76         const a = placeholders(enE[key]);
77         const b = placeholders(locE[key]);
78         if (a.join(",") !== b.join(",")) mismatches.push({ key, en: a, got: b });
79     }
80     expect(mismatches).toEqual([]);
81 });
82 }
83 });
84
85 describe("es 3-form plural (CLDR one/many/other) ? MULTILINGUAL Phase A", () => {
86     // Spanish cardinal is {one, many, other}; every es ICU plural message MUST carry a
87     // (the `many` category fires on large/compact numbers) or intl-messageformat silently
88     // These are the plural-bearing keys in the catalog.
89     const es = JSON.parse(readFileSync(join(LOCALES_DIR, "es", "common.json"), "utf8")) as
Record<string, Record<string, string>>;
90     const pluralKeys: Array<[string, string]> = [
91         ["footer", "count"],
92         ["footer", "floorSomeLead"],
93         ["selection", "count"],
94     ];
95     for (const [ns, key] of pluralKeys) {
96         it(`es ${ns}.${key} carries one/many/other arms`, () => {
97             const msg = es[ns][key];
98             expect(msg, `${ns}.${key}`).toMatch(/\\bone\\s*\\{/);
99             expect(msg, `${ns}.${key}`).toMatch(/\\bmany\\s*\\{/);
100             expect(msg, `${ns}.${key}`).toMatch(/\\bother\\s*\\{/);
101         });
102     }
103 });
104
105 describe("no-bare-strings ? user-facing component attributes go through t()", () => {
106     const FILES = [
107         "App.tsx",
108         "components/QueryBar.tsx",
109         "components/TapFilters.tsx",
110         "components/RallyGrid.tsx",
111         "components/SelectionFooter.tsx",
112         "components/CompileResult.tsx",
113         "components/LanguageSwitcher.tsx",
114         "components/IngestPanel.tsx",
115     ];
116     // A hardcoded placeholder / aria-label / title with letters is a bare-string
117     // must be an expression ({t(...)}), not a quoted literal.
118     const BARE_ATTR = /\\b(?:placeholder|aria-label|title)\\s*=\\s*"\\^[^"]*[A-Za-z]{2}\\^[^"]*"\\/g;
119
120     for (const f of FILES) {
121         it(`${f}: no hardcoded placeholder/aria-label/title`, () => {
122             const src = readFileSync(join(SRC_DIR, f), "utf8");
123             expect(src.match(BARE_ATTR) ?? []).toEqual([]);
124         });
125     }
126 });
127
128 describe("Indic fonts (B4) ? self-hosted OFL Noto, no runtime CDN", () => {
129     it("main.tsx bundles the OFL Noto Kannada + Devanagari fonts (self-hosted, not Google
Fonts)", () => {
130         const main = readFileSync(join(SRC_DIR, "main.tsx"), "utf8");
131         expect(main).toMatch(/@fontsource-variable\\noto-sans-kannada\\/);
132         expect(main).toMatch(/@fontsource-variable\\noto-sans-devanagari\\/);
133     });
134     it("the font stack falls back to Noto for Kannada + Devanagari glyphs", () => {
135         const css = readFileSync(join(SRC_DIR, "index.css"), "utf8");

```

```

136     expect(css).toMatch(/Noto Sans Kannada Variable/);
137     expect(css).toMatch(/Noto Sans Devanagari Variable/);
138   });
139 });
140
141 // The Studio SPA must render correctly on an AIR-GAPPED appliance (GPU/CPU/RAM, no
internet) ?
142 // every asset (fonts, styles, scripts) is bundled local, nothing pulled from a
CDN/Google Fonts.
143 describe("offline / self-contained ? no external runtime assets", () => {
144   const CDN = /googleapis|gstatic|fonts\.google|cdnjs|unpkg|jsdelivr|bcdn\.[a-z]/i;
145   const FILES = [
146     "../index.html", // web/index.html
147     "main.tsx",
148     "index.css",
149   ];
150   for (const rel of FILES) {
151     it(`${rel}: no external font/CDN reference`, () => {
152       expect(readFileSync(join(SRC_DIR, rel), "utf8")).not.toMatch(CDN);
153     });
154   }
155   it("web/index.html loads no absolute-URL <link>/<script src> (everything is
bundled/local)", () => {
156     const html = readFileSync(join(SRC_DIR, "..", "index.html"), "utf8");
157     expect(html).not.toMatch(/(?:href|src)\s*=\s*["']https?:\/\//i);
158   });
159 });
160

```

16. user (2026-06-21T19:02:28.141Z)

```

hygiene.test.ts
i18n.test.tsx
index.ts
---render tests---
src/api/client.test.ts
src/api/locale.test.ts
src/api/upload.test.ts
src/components/a4.touch.test.tsx
src/components/a8.status.test.tsx
src/components/build-error.test.tsx
src/components/IngestPanel.test.tsx
src/components/QueryBar.examples.test.ts
src/components/TapFilters.test.tsx
src/components/upload.test.tsx
src/hooks/useIngestJob.test.ts
src/hooks/useUpload.test.ts
src/i18n/hygiene.test.ts
src/i18n/i18n.test.tsx
src/lib/errors.test.ts
src/lib/errorText.test.ts
src/lib/floor.test.ts
src/lib/format.test.ts
src/lib/query.test.ts
src/lib/selection.test.ts
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

17. assistant (2026-06-21T19:02:31.561Z)

Now let me check the i18n render test and search for any test that asserts specific kn/hi string content (which my edits could break).

18. user (2026-06-21T19:02:32.517Z)

```

1      // @vitest-environment jsdom
2      //
3      // The SPA had NO render tests before this (verified) ? this file adds the jsdom safety
net for the
4      // i18n scaffold (tracker B1): real copy renders from the catalog, the switcher persists
the choice,
5      // missing-locale falls back to English, and ICU plurals format. It mounts the real <App
/>.
6      import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
7      import { render, screen, cleanup, fireEvent } from "@testing-library/react";
8      import i18n, { LANG_STORAGE_KEY } from "../index";
9      import App from "../App";
10
11      beforeEach(async () => {
12          // No engine in the test env: stub fetch so App's getConfig()/listRallies effects
reject quietly
13          // (App handles this gracefully ? cfg stays null) instead of hitting the network.
14          vi.stubGlobal("fetch", vi.fn(() => Promise.reject(new Error("no engine in test"))));
15          window.localStorage.clear();
16          await i18n.changeLanguage("en"); // also guarantees init completed before the first
render
17      });
18
19      afterEach(() => {
20          cleanup();
21          vi.unstubAllGlobals();
22      });
23
24      describe("i18n scaffold (B1)", () => {
25          it("renders user-facing copy from the en catalog (not raw keys)", () => {
26              render(<App />);
27              expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("Pick the rallies
for your reel");
28              expect(screen.getByRole("button", { name: "Select all" })).toBeTruthy();
29          });
30
31          it("B2: the extracted component strings render (no leftover raw keys across the
page)", () => {
32              const { container } = render(<App />);
33              expect(screen.getByRole("button", { name: "Apply" })).toBeTruthy(); // QueryBar
34              expect(screen.getByRole("button", { name: /Build reel/ })).toBeTruthy(); //
SelectionFooter
35              expect(container.textContent).toContain("rally length"); // <Trans> hero intro +
query support
36              expect(container.textContent).toMatch(/\d+ rallies selected \. \d+:\d+ total/); //
ICU footer summary
37              expect(container.textContent).toMatch(/Rally \d+/); // RallyGrid card
38              // No untranslated catalog keys leaked into the DOM.
39              expect(container.textContent).not.toMatch(/\b(app|grid|footer|query|mode|result|buil
d)\.[a-zA-Z]+\./);
40          });
41
42          it("offers a language switcher and persists the choice (persisted pref wins on next
visit)", () => {
43              render(<App />);
44              const knBtn = screen.getByRole("button", { name: "?????" });
45              fireEvent.click(knBtn);
46              expect(i18n.language).toBe("kn");
47              expect(window.localStorage.getItem(LANG_STORAGE_KEY)).toBe("kn");
48              expect(knBtn.getAttribute("aria-pressed")).toBe("true"); // highlights the chosen
locale
49              // Switching actually re-renders the page into Kannada (the kn catalog, B5).
50              expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("????? ????????
????????????? ??????");
51          });

```

```

52
53     it("B5: switching to Kannada renders the kn catalog (Kannada copy, not English)",
async () => {
54         await i18n.changeLanguage("kn");
55         render(<App />);
56         expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("????? ??????????
????????????? ??????");
57         expect(screen.getByRole("button", { name: "?????????" })).toBeTruthy(); // Apply ?
Kannada
58         expect(screen.getByRole("button", { name: "????????? ??????????" })).toBeTruthy(); //
Select all ? Kannada
59         expect(document.documentElement.lang).toBe("kn"); // <html lang> tracks the active
locale (a11y)
60     });
61
62     it("B6: switching to Hindi renders the hi catalog (extensibility proof ? just a
catalog)", async () => {
63         await i18n.changeLanguage("hi");
64         render(<App />);
65         expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("???? ???? ?? ???
???????? ??????");
66         expect(screen.getByRole("button", { name: "???? ??????" })).toBeTruthy(); // Apply ?
Hindi
67         expect(document.documentElement.lang).toBe("hi");
68     });
69
70     it("a locale with no catalog falls back to English (never a raw key)", async () => {
71         await i18n.changeLanguage("ta"); // ta has no catalog
72         render(<App />);
73         expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("Pick the rallies
for your reel");
74     });
75
76     it("formats ICU plurals from the catalog", () => {
77         expect(i18n.t("selection.count", { count: 1 })).toBe("1 rally selected");
78         expect(i18n.t("selection.count", { count: 3 })).toBe("3 rallies selected");
79     });
80 });
81
82 // The owner's bar (#41): a user whose BROWSER/OS language is Kannada/Hindi must auto-
get the
83 // respective version. These prove the negotiation NOW at the detection level (B1, en-
only); the
84 // kn/hi CONTENT assertions ride the catalogs in B5/B6. See FRIENDLY_AND_MULTILINGUAL.md
$6.5 (N1?N9).
85 describe("browser-language negotiation (B1: N1/N3?N7)", () => {
86     const detectBases = () => {
87         const d = i18n.services.languageDetector.detect();
88         return (Array.isArray(d) ? d : [d]).filter(Boolean).map((l) =>
String(l).split("-")[0]);
89     };
90
91     it("N1/N5: a Kannada browser (kn-IN) is auto-detected as kn for a first-time visitor",
() => {
92         window.localStorage.clear(); // no persisted preference yet
93         vi.stubGlobal("navigator", { language: "kn-IN", languages: ["kn-IN", "en-IN"] });
94         expect(detectBases()).toContain("kn"); // the kn-IN region subtag normalizes to base
kn
95     });
96
97     it("N2: a Hindi browser (hi-IN) is auto-detected as hi (now a supported locale)", ()
=> {
98         window.localStorage.clear();
99         vi.stubGlobal("navigator", { language: "hi-IN", languages: ["hi-IN", "en-IN"] });
100        expect(detectBases()).toContain("hi");

```

```

101     });
102
103     it("N4: a persisted preference beats the browser language (returning user keeps their
choice)", () => {
104         window.localStorage.setItem(LANG_STORAGE_KEY, "kn");
105         vi.stubGlobal("navigator", { language: "en-US", languages: ["en-US"] });
106         expect(detectBases()[0]).toBe("kn");
107     });
108
109     it("N3: persisted pref and ?lang are negotiated BEFORE the browser language", () => {
110         const order = i18n.options.detection?.order ?? [];
111         expect(order.indexOf("localStorage")).toBeGreaterThanEqual(0);
112         expect(order.indexOf("localStorage")).toBeLessThan(order.indexOf("navigator"));
113         expect(order.indexOf("querystring")).toBeLessThan(order.indexOf("navigator"));
114     });
115
116     it("N6: an unsupported browser language (Tamil) serves English content ? never a
locale we can't render", async () => {
117         window.localStorage.clear();
118         await i18n.changeLanguage("ta"); // ta is not in SUPPORTED_LOCALES
119         render(<App />);
120         expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("Pick the rallies
for your reel");
121     });
122
123     it("N7: a key with no catalog entry falls back to English (no raw key)", async () => {
124         await i18n.changeLanguage("ta"); // ta has no catalog ? keys resolve via
fallbackLng=en
125         expect(i18n.t("app.title")).toBe("Pick the rallies for your reel");
126         expect(i18n.t("bulk.selectAll")).toBe("Select all");
127     });
128 });
129
130 // MULTILINGUAL_EXPANSION Phase A ? es/da/id render + auto-negotiate, same rigor as
en/kn/hi.
131 // es is the only authoring wrinkle (CLDR 3-form {one, many, other}); da is 2-form; id
is single-form.
132 describe("MULTILINGUAL Phase A (es/da/id)", () => {
133     it("renders each new locale's catalog (real translations, not English / not raw
keys)", async () => {
134         const titles: Record<string, string> = {
135             es: "Elige las jugadas para tu reel",
136             da: "Vælg duellerne til din reel",
137             id: "Pilih reli untuk reel kamu",
138         };
139         for (const [loc, title] of Object.entries(titles)) {
140             await i18n.changeLanguage(loc);
141             cleanup();
142             const { container } = render(<App />);
143             expect(screen.getByRole("heading", { level: 1 }).textContent, loc).toBe(title);
144             expect(container.textContent,
loc).not.toMatch(/\b(app|grid|footer|query|ingest)\.[a-zA-Z]+/); // no raw keys
145         }
146         await i18n.changeLanguage("en");
147     });
148
149     it("es uses the 3-form plural (singular vs plural word forms)", async () => {
150         await i18n.changeLanguage("es");
151         expect(i18n.t("selection.count", { count: 1 })).toBe("1 jugada seleccionada"); //
one
152         expect(i18n.t("selection.count", { count: 2 })).toBe("2 jugadas seleccionadas"); //
other
153         await i18n.changeLanguage("en");
154     });
155

```

```

156     it("id is single-form (only an 'other' arm) and still interpolates the count", async
157     () => {
158         await i18n.changeLanguage("id");
159         expect(i18n.t("selection.count", { count: 1 })).toBe("1 reli dipilih");
160         expect(i18n.t("selection.count", { count: 5 })).toBe("5 reli dipilih");
161         await i18n.changeLanguage("en");
162     });
163     it("auto-negotiates es/da/id from the browser language (region subtags normalize to
164     base)", () => {
165         for (const [lng, base] of [["es-ES", "es"], ["da-DK", "da"], ["id-ID", "id"]] as
166         const) {
167             window.localStorage.clear();
168             vi.stubGlobal("navigator", { language: lng, languages: [lng] });
169             const d = i18n.services.languageDetector.detect();
170             const bases = (Array.isArray(d) ? d : [d]).filter(Boolean).map((l) =>
171             String(l).split("-")[0]);
172             expect(bases, lng).toContain(base);
173         }
174     });
175 }

```

19. user (2026-06-21T19:02:36.211Z)

```

src/api/client.test.ts:91:    vi.stubGlobal("fetch", mockFetch(200, { stitching: {
min_shot_count: 4 } }));
src/components/build-error.test.tsx:66:    vi.stubGlobal("fetch", mockFetch(async () =>
httpErr(400, JSON.stringify({ detail: "rally below min_shot_count" })));
src/components/build-error.test.tsx:68:    expect(await screen.findByText(/rally below
min_shot_count/i)).toBeTruthy();
src/lib/errorText.test.ts:18:    expect(errorText(i18n.t, { code: "engine", detail:
"min_shot_count too low" })).toBe("min_shot_count too low");
src/lib/errorText.test.ts:42:    expect(errorText(i18n.t, { code: "networkError"
})).toBe("????? ??????????????. ??? ??????????????");
---done grep tests---
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

20. assistant (2026-06-21T19:02:47.733Z)

Key finding: [errorText.test.ts:42](web/src/lib/errorText.test.ts:42) asserts the exact kn
`networkError` string ? I'll need to update that test when I change it. Let me read that test
file.

21. user (2026-06-21T19:02:48.065Z)

```

1      // @vitest-environment jsdom
2      //
3      // errorText is the single mapping from a classified error ? user-facing text, shared by
4      the ingest
5      // and build flows (A5). These pin: the engine message passes through verbatim, a
6      detail-less engine
7      // error degrades to a friendly localized line (never a raw key), and every non-engine
8      code resolves
9      // to a real catalog string in the shared `errors.*` namespace.
10     import { describe, it, expect, beforeAll } from "vitest";
11     import i18n from "../i18n";
12     import { errorText } from "../errorText";
13     import type { IngestErrorCode } from "../errors";
14
15     beforeAll(async () => {
16         await i18n.changeLanguage("en");
17     });
18 }

```


29. user (2026-06-21T19:08:43.470Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

30. user (2026-06-21T19:08:49.675Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

31. user (2026-06-21T19:08:53.125Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

32. user (2026-06-21T19:09:02.715Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

33. user (2026-06-21T19:09:03.721Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

34. assistant (2026-06-21T19:09:06.232Z)

Now the Hindi `ingest.\*` and `errors.\*` edits:

35. user (2026-06-21T19:09:09.560Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

36. user (2026-06-21T19:09:12.387Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

37. user (2026-06-21T19:09:14.743Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

38. user (2026-06-21T19:09:17.589Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

39. user (2026-06-21T19:09:20.037Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/hi/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

40. assistant (2026-06-21T19:09:24.206Z)

Hindi done. Now the Kannada catalog:

41. user (2026-06-21T19:09:26.553Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

42. user (2026-06-21T19:09:45.944Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

43. user (2026-06-21T19:09:47.677Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

44. assistant (2026-06-21T19:09:51.656Z)

Now the Kannada `ingest.\*` and `errors.\*` edits:

45. user (2026-06-21T19:10:00.848Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

46. user (2026-06-21T19:10:05.874Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

47. user (2026-06-21T19:10:13.321Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

48. user (2026-06-21T19:10:17.625Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

49. user (2026-06-21T19:10:23.552Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/locales/kn/common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

50. assistant (2026-06-21T19:10:26.595Z)

Now update the test that pins the kn `networkError` string:

51. user (2026-06-21T19:10:29.614Z)

The file C:/Users/avidu/Projects/khelsutra-i18n-review/web/src/lib/errorText.test.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

52. assistant (2026-06-21T19:10:46.923Z)

The PostToolUse hook mentions a preview server, but browser preview can't meaningfully verify translation *quality* (that needs human reading of Kannada/Hindi) ? the user prescribed the test suite as the verification path, which is the right tool here. First, a critical check: I edited `/locales/`, but `site/` ships a **committed mirror** under `site/public/locales/` guarded by a drift test. If the site mirror includes these namespaces, CI's site job will fail unless I re-sync it. Let me verify.

53. user (2026-06-21T19:10:47.891Z)

```
1      // DRIFT GUARD for the one build step the otherwise no-build site has (tracker B3 /
D-CATALOG-HOME):
2      // the catalogs the marketing site ships (site/public/locales/) MUST be byte-identical
to the single
```

```

3      // source of truth (repo-root /locales/). If someone edits /locales without re-running
the copy, this
4      // fails CI with the exact fix. Run `npm run copy:locales` to resync.
5      import { test } from "node:test";
6      import assert from "node:assert/strict";
7      import { readFileSync, readdirSync } from "node:fs";
8      import { join } from "node:path";
9
10     const SRC = join(import.meta.dirname, "..", "..", "locales");
11     const PUB = join(import.meta.dirname, "..", "public", "locales");
12
13     test("site/public/locales mirrors repo-root /locales exactly (else: npm run
copy:locales)", () => {
14         const locales = readdirSync(SRC, { withFileTypes: true }).filter((d) =>
d.isDirectory()).map((d) => d.name);
15         assert.ok(locales.length >= 3, "expected at least en/kn/hi catalogs in /locales");
16
17         const shipped = new Set(readdirSync(PUB, { withFileTypes: true }).filter((d) =>
d.isDirectory()).map((d) => d.name));
18         for (const loc of locales) {
19             const src = readFileSync(join(SRC, loc, "common.json"), "utf8");
20             assert.ok(shipped.has(loc), `site/public/locales/${loc} is missing ? run npm run
copy:locales`);
21             const pub = readFileSync(join(PUB, loc, "common.json"), "utf8");
22             assert.equal(pub, src, `site/public/locales/${loc}/common.json drifted from
/locales/${loc} ? run npm run copy:locales`);
23         }
24         // No stale extra locales shipped that aren't in the source of truth.
25         for (const loc of shipped) {
26             assert.ok(locales.includes(loc), `site/public/locales/${loc} has no source in
/locales ? run npm run copy:locales`);
27         }
28     });
29

```

54. user (2026-06-21T19:10:50.770Z)

```

=== copy-locales script ===
// Mirror the single-source-of-truth catalogs (repo-root /locales, D-CATALOG-HOME) into the
site's
// static public/ dir, so the dependency-free marketing site can fetch them at runtime via the
// t()/data-i18n shim (public/i18n.js). This copy is the ONLY build step the otherwise no-build
site
// has. Run on demand (`npm run copy:locales`) or automatically before deploy (`predeploy`).
//
// The committed copy under site/public/locales/ is what actually ships (static hosts serve it
with
// no build); the drift-guard test (src/locales-drift.test.js) fails CI if it goes stale vs
/locales.
import { cpSync, rmSync, mkdirSync, readdirSync } from "node:fs";
import { join } from "node:path";

const SRC = join(import.meta.dirname, "..", "..", "locales");
const DEST = join(import.meta.dirname, "..", "public", "locales");

rmSync(DEST, { recursive: true, force: true });
mkdirSync(DEST, { recursive: true });
cpSync(SRC, DEST, { recursive: true });

console.log(`copy-locales: ${readdirSync(DEST).join(", ")} -> site/public/locales/`);
=== site committed kn copy: does it contain the console keys? ===
9
=== package.json scripts (site) ===
6:   "test": "node --test",
7:   "test:cf": "node --test src/index.test.js",

```

```
8:   "test:vm": "node --test src/server.test.js",
9:   "test:deploy": "node --test src/deploy-config.test.js src/assets.test.js",
10:   "copy:locales": "node scripts/copy-locales.mjs",
11:   "predeploy": "node scripts/copy-locales.mjs",
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

55. assistant (2026-06-21T19:10:58.072Z)

Confirmed ? the drift guard requires the site mirror to be **byte-identical** to ``/locales``, and the committed site copy is a full mirror (includes the console keys). So I must re-run ``copy:locales`` to resync ``site/public/locales/{kn,hi}/common.json``, or the site CI job fails. The copy script uses only Node built-ins (no deps). Running it now:

56. user (2026-06-21T19:11:00.630Z)

```
> copy:locales
> node scripts/copy-locales.mjs

copy-locales: da, en, es, hi, id, kn -> site/public/locales/
---drift test---
? site/public/locales mirrors repo-root /locales exactly (else: npm run copy:locales) (5.908ms)
? tests 1
? suites 0
? pass 1
? fail 0
? cancelled 0
? skipped 0
? todo 0
? duration_ms 103.2215
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

57. assistant (2026-06-21T19:11:04.564Z)

Drift guard green. Now let me verify the changeset and JSON validity, then run the prescribed web suite (``npm ci && typecheck && test``).

58. user (2026-06-21T19:11:07.556Z)

```
=== changed files ===
M locales/hi/common.json
M locales/kn/common.json
M site/public/locales/hi/common.json
M site/public/locales/kn/common.json
M web/src/lib/errorText.test.ts
=== JSON validity (all 6 + site mirrors) ===
OK locales/en/common.json
OK locales/kn/common.json
OK locales/hi/common.json
OK locales/es/common.json
OK locales/da/common.json
OK locales/id/common.json
OK site/public/locales/kn/common.json
OK site/public/locales/hi/common.json
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

59. assistant (2026-06-21T19:11:11.811Z)

All JSON valid; changeset is clean (2 source catalogs + 2 regenerated site mirrors + 1 test pin). Now the prescribed web verification ? ``npm ci && typecheck && test``. This will take a bit:

60. user (2026-06-21T19:11:18.397Z)

added 180 packages, and audited 181 packages in 5s

```
34 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

61. user (2026-06-21T19:11:25.488Z)

```
=== TYPECHECK ===
```

```
> khelsutra-web@0.0.0 typecheck
> tsc --noEmit
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

62. assistant (2026-06-21T19:11:27.942Z)

```
Typecheck clean. Now the full test suite:
```

63. user (2026-06-21T19:11:36.456Z)

```
stdout | src/components/build-error.test.tsx > A5: friendly build-error
mapping > LW2: the active UI locale is sent to /api/compile (drives the localized
reel watermark)
{"ts":"2026-06-21T19:11:36.049Z","level":"info","event":"build.start","cid":"08ae87b6-
4a6c-4869-8c9b-
7a4f776f74ab","video_id":"vidX","filename":"highlights.mp4","selected":2,"total_sec":25,"below_f
loor":0}
{"ts":"2026-06-21T19:11:36.049Z","level":"info","event":"api.request","cid":"08ae87b6-4a6c-4869-
8c9b-7a4f776f74ab","method":"POST","url":"/api/compile"}
```

```
stdout | src/components/build-error.test.tsx > A5: friendly build-error
mapping > LW2: the active UI locale is sent to /api/compile (drives the localized
reel watermark)
{"ts":"2026-06-
21T19:11:36.058Z","level":"info","event":"api.response","cid":"08ae87b6-4a6c-4869-8c9b-
7a4f776f74ab","method":"POST","url":"/api/compile","status":200}
```

```
stdout | src/components/build-error.test.tsx > A5: friendly build-error
mapping > LW2: the active UI locale is sent to /api/compile (drives the localized
reel watermark)
{"ts":"2026-06-21T19:11:36.058Z","level":"info","event":"build.done","cid":"08ae87b6-
4a6c-4869-8c9b-
7a4f776f74ab","video_id":"vidX","filename":"highlights.mp4","filepath":"/o/r.mp4","url":"/api/hi
ghlights/vidX/highlights.mp4"}
```

```
? src/components/build-error.test.tsx (4 tests)
```

```
932ms
```

```
? A5: friendly build-error mapping > a network / engine-down
compile failure shows the friendly 'is it running?' line 593ms
stdout | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.056Z","level":"info","event":"api.request","cid":"5a77eb11-
ccb5-4267-b44e-f5f9a2164516","method":"GET","url":"/api/config"}
```

```
stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-
21T19:11:36.073Z","level":"error","event":"api.network_error","cid":"5a77eb11-ccb5-4267-b44e-
f5f9a2164516","method":"GET","url":"/api/config","error":"Error: no engine in test"}
```

```
stdout | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.090Z","level":"info","event":"api.request","cid":"e86a2baa-
8357-4ac3-a09d-e169a2894619","method":"GET","url":"/api/config"}
```

```

stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.111Z","level":"error","event":"api.network_error","cid":"e86a2baa-8357-4ac3-a09d-e169a2894619","method":"GET","url":"/api/config","error":"Error: no engine in test"}

stdout | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.125Z","level":"info","event":"api.request","cid":"2e9ae718-b4b2-47d8-970b-cc07a1fab256","method":"GET","url":"/api/config"}

stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.141Z","level":"error","event":"api.network_error","cid":"2e9ae718-b4b2-47d8-970b-cc07a1fab256","method":"GET","url":"/api/config","error":"Error: no engine in test"}
{"ts":"2026-06-21T19:11:36.142Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.142Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-21T19:11:36.142Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}

? src/i18n/i18n.test.tsx (17 tests)
1007ms
? i18n scaffold (B1) > renders user-facing copy from the en
catalog (not raw keys) 492ms

Test Files 20 passed (20)
Tests 177 passed (177)
Start at 00:41:30
Duration 5.73s (transform 1.49s, setup 478ms, collect 9.13s, tests 4.58s,
environment 26.60s, prepare 4.93s)
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

64. assistant (2026-06-21T19:11:45.609Z)

All 177 web tests pass (20 files) ? hygiene parity, es 3-form, the 17 i18n render tests, and the updated `errorText` kn pin all green. Since CI also runs the **site** job, let me run the site suite too (the drift guard plus the rest):

65. user (2026-06-21T19:11:48.647Z)

```

Node.js v26.1.0
? src\ux.dom.test.js (113.848ms)
? tests 37
? suites 0
? pass 34
? fail 3
? cancelled 0
? skipped 0
? todo 0
? duration_ms 614.4224

? failing tests:

test at src\assets.test.js:1:1

```

```
? src\assets.test.js (122.9605ms)
  'test failed'

test at src\i18n.shim.test.js:1:1
? src\i18n.shim.test.js (140.7282ms)
  'test failed'

test at src\ux.dom.test.js:1:1
? src\ux.dom.test.js (113.848ms)
  'test failed'
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

66. assistant (2026-06-21T19:11:52.581Z)

3 site tests failing. I need to see the actual errors and determine whether they're caused by my change or pre-existing. Let me get details:

67. user (2026-06-21T19:11:56.249Z)

```
Error [ERR_MODULE_NOT_FOUND]: Cannot find package 'jsdom' imported from
C:\Users\avidu\Projects\khelsutra-i18n-review\site\src\i18n.shim.test.js
    at Object.getPackageJSONURL (node:internal/modules/package_json_reader:301:9)
    at packageResolve (node:internal/modules/esm/resolve:768:81)
    at moduleResolve (node:internal/modules/esm/resolve:859:18)
    at defaultResolve (node:internal/modules/esm/resolve:992:11)
    at #cachedDefaultResolve (node:internal/modules/esm/loader:694:20)
    at #resolveAndMaybeBlockOnLoaderThread (node:internal/modules/esm/loader:711:38)
    at ModuleLoader.resolveSync (node:internal/modules/esm/loader:743:52)
    at #resolve (node:internal/modules/esm/loader:676:17)
    at ModuleLoader.getOrCreateModuleJob (node:internal/modules/esm/loader:596:35)
    at ModuleJob.syncLink (node:internal/modules/esm/module_job:163:33) {
  code: 'ERR_MODULE_NOT_FOUND'
}
```

```
Node.js v26.1.0
? src\i18n.shim.test.js (82.1783ms)
? tests 1
? suites 0
? pass 0
? fail 1
? cancelled 0
? skipped 0
? todo 0
? duration_ms 90.0526
```

? failing tests:

```
test at src\i18n.shim.test.js:1:1
? src\i18n.shim.test.js (82.1783ms)
  'test failed'
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

68. assistant (2026-06-21T19:12:04.970Z)

Those 3 failures are `jsdom` module-not-found ? site dev-deps aren't installed (I only ran `npm ci` in `web`). That's an environment gap, not my change. Let me install site deps and re-run:

69. user (2026-06-21T19:12:13.339Z)

```
8 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
=== full site test run ===
```

```
? [VM host · N8] non-localized assets are NOT rewritten (only the 3 marketing pages negotiate)
(1.9497ms)
? [VM host] serves the shared catalogs the shim fetches (/locales/*/common.json as JSON)
(2.5447ms)
? submit shows an obvious in-flight state (disabled + aria-busy + 'Sending?') (478.934ms)
? on failure the button is restored ? never stuck on a frozen 'Sending?' (91.0557ms)
? after sign-up BOTH videos appear and a ? closes the reveal (69.2043ms)
? the auto-close HOLDS while the reel is playing (no mid-watch cut-off) (62.5917ms)
? tests 68
? suites 0
? pass 68
? fail 0
? cancelled 0
? skipped 0
? todo 0
? duration_ms 2604.4972
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

70. assistant (2026-06-21T19:12:19.323Z)

All 68 site tests pass now (including the drift guard) ? the earlier 3 failures were just the missing `jsdom` dev-dep. Let me eyeball the final diff of the source catalogs before committing:

71. user (2026-06-21T19:12:20.789Z)

```
locales/hi/common.json          | 32 ++++++-----
locales/kn/common.json          | 22 ++++++-----
site/public/locales/hi/common.json | 32 ++++++-----
site/public/locales/kn/common.json | 22 ++++++-----
web/src/lib/errorText.test.ts    | 2 +-
5 files changed, 55 insertions(+), 55 deletions(-)
===== SOURCE CATALOG DIFFS (kn + hi) =====
diff --git a/locales/hi/common.json b/locales/hi/common.json
index d9d42f0..4a94a58 100644
--- a/locales/hi/common.json
+++ b/locales/hi/common.json
@@ -1,8 +1,8 @@
 {
   "app": {
-    "badge": "????-???? ??. . ??????",
+    "badge": "???? ?????? ?????? . ??????",
     "title": "???? ???? ???? ??????? ??????",
-    "intro": "<b>???? ????</b> ???? ???? ???? ???? ???? ???? ???? 15?</i> ?
??? ????? ???? ???? ?????? ???? ?????? ???? ???? <b>????</b> ???? ???? ????; ???? ???? ?????
?????? ???? ????"
+    "intro": "<b>???? ????</b> ???? ???? ???? ?????? ???? ???? <i>????? ???? ???? 15?</i> ?
??? ????? ???? ???? ?????? ???? ?????? ???? ???? <b>????</b> ???? ????; ???? ???? ?????
?????? ???? ????"
   },
   "lang": {
     "label": "????",
@@ -34,7 +34,7 @@
     "unreadable": "?? ??? ???? ???? ? ?over 10s? ?? ?top 15 longest? ????????",
     "hint": "10 ????? ???? ?????????? ???? ???? over 10s ???? ???? ? ???? ???? ????
?????",
     "try": "????? ??????:",
-    "support": "<b>???? ????</b> ???? ???? ???? ?????? ???? ???? ???? ?????? ???? ???-?????
<x>????????????</x> ?? (?????????)?"
+    "support": "<b>???? ????</b> ???? ???? ???? ?????? ???? ???? ???? ???? ???? ????-?????
<x>????????</x> ?? (??? ???? ???? ????)"
   },
   "grid": {
     "empty": "????? ???? ???? ???? ???? ???? ???? ????",
@@ -43,9 +43,9 @@
     "notInReel": "???? {n}, ??? ??? ?????",
```


[illegible]

[illegible]

@@ -72,28 +72,28 @@

```
    },
    "ingest": {
      "title": "????? ??????????????",
      - "intro": "???????????????? ???? ??????? ? ????? ???? ???? ?????????????? ???? ?
        ??? ?????????????? ?????????????? ?????? ?????? ?????????????? ??????????????",
      + "intro": "KhelSutra ?????? ?????? ?????????? ??????, ??? ?????? ?????????????? ?????????????? ?
        ?????? ?????? ?????????? ??????????????",
      "uriPlaceholder": "gs://your-bucket/match.mp4",
      "uriAriaLabel": "????? ???? ? gs:// ???? URL ???? ?????? ?????????????????? ??",
      - "hint": "gs:// ???? ?????, ??? ?????? ?????????????? ?????? ?????? ?????? ??????",
      + "hint": "?????-???????? ???? (gs://?), ??? KhelSutra ?????????????? ?????????????? ?????
        ?????? ??????",
      "submit": "????? ??????????????",
      "submitting": "?????????????????",
      "cancel": "?????????????",
      "working": "????? ?????? ??????????????????????????",
      "progress": "{progress}",
      "retry": "????? ??????????????",
      - "reassure": "????????? ?????? ? ?????? ?????? ?????????????????? ?????????? ?????????????? ????
        ?????????????????; ?????? ?????????? ?????? ?????????? ??????????????",
      + "reassure": "????????? ?????? ? ?????? ?????? ?????????????????? KhelSutra ???
        ?????????????????; ?????? ?????????? ?????? ?????????? ??????????????",
      "statusFailed": "?????????????????????????",
      "chooseFile": "????? ???? ??????",
      "uploadAria": "????????? ?????? ?????? ???? ??????",
      - "fileHint": "? ?????????? .mp4 ???? .mov ?????? ? ??? ?????????? ?????????, ???? ?????????
        ?????????????? ??????????????",
      + "fileHint": "? ?????????? .mp4 ???? .mov ?????? ? ??? ?????????? ?????????, ??? KhelSutra
        ?????????????? ??????????????",
      "orLink": "????? ?????? ?????? ??????",
      "uploadingStatus": "????????? ??????????????",
      "uploadProgress": "????????? ??????"
    },
    "errors": {
      - "networkError": "????? ?????????????????? ??? ??????????????????",
      - "invalidUri": "????? ?????? ?????????? ?????? ? gs:// ???? ???? ?????????????????? ??.",
      + "networkError": "KhelSutra ?????????????????? ??? ??????????????????",
      + "invalidUri": "????? ?????? ?????????? ?????? ? gs:// ???? ???? KhelSutra ?????????????????
        ?????? ??.",
      "emptyResult": "????????? ?????????? ??? ? ?????????????? ?????? ?????????? ?????????? ?????????
        ?????? ?????????????? ??????????????",
      "timeout": "????????????????? ?????? ??? ?????????????????????? ??? ?????? ?????????????????? ?
        ?????? ?????????????????? ?????????? ???, ??? ?????? ??????????????",
      "engineGeneric": "??? ?????????? ?????????? ?????? ??????????"
    }
  },
  "errors": {
    - "networkError": "????? ?????????????????? ??? ??????????????????",
    - "invalidUri": "????? ?????? ?????????? ?????? ? gs:// ???? ???? ?????????????????? ??.",
    + "networkError": "KhelSutra ?????????????????? ??? ??????????????????",
    + "invalidUri": "????? ?????? ?????????? ?????? ? gs:// ???? ???? KhelSutra ?????????????????
      ?????? ??.",
    "emptyResult": "????????? ?????????? ??? ? ?????????????? ?????? ?????????? ?????????? ?????????
      ?????? ?????????????? ??????????????",
    "timeout": "????????????????? ?????? ??? ?????????????????????? ??? ?????? ?????????????????? ?
      ?????? ?????????????????? ?????????? ???, ??? ?????? ??????????????",
    "engineGeneric": "??? ?????????? ?????????? ?????? ??????????"
  }
},
"errors": {
  - "networkError": "????? ?????????????????? ??? ??????????????????",
  - "invalidUri": "????? ?????? ?????????? ?????? ? gs:// ???? ???? ?????????????????? ??.",
  + "networkError": "KhelSutra ?????????????????? ??? ??????????????????",
  + "invalidUri": "????? ?????? ?????????? ?????? ? gs:// ???? ???? KhelSutra ?????????????????
    ?????? ??.",
  "emptyResult": "????????? ?????????? ??? ? ?????????????? ?????? ?????????? ?????????? ?????????
    ?????? ?????????????? ??????????????",
  "timeout": "????????????????? ?????? ??? ?????????????????????? ??? ?????? ?????????????????? ?
    ?????? ?????????????????? ?????????? ???, ??? ?????? ??????????????",
  "engineGeneric": "??? ?????????? ?????????? ?????? ??????????"
}
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

72. assistant (2026-06-21T19:12:38.155Z)

The diff is exactly as intended: placeholders all preserved, brand is Latin `KhelSutra`, engine/server jargon dropped where en avoids it, the ``min_shot_count`` token removed, fluency fixes applied ? and `uriAriaLabel` correctly **kept** "engine host"/"bucket URL" (en uses them there).

Now let me commit. First, re-verify branch safety (master hasn't moved under me; only my work on top) and stage explicit paths:

73. user (2026-06-21T19:12:42.562Z)

```
=== branch ===
fix/kn-hi-spa-console-native-review
=== HEAD ===
1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)
=== commits on top of origin/master (should be empty pre-commit) ===
```

```
=== is origin/master ancestor of HEAD? ===
NO-master-moved
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

74. assistant (2026-06-21T19:12:50.038Z)

Master moved under me (a sibling merged past `1cd84af`). Expected churn. Let me see the new `origin/master` and whether it touched the catalogs (conflict risk) before I commit and rebase onto it:

75. user (2026-06-21T19:12:52.889Z)

```
=== new origin/master ===
aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)
1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics) (#73)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-HOME) (#70)
=== files changed since my base (1cd84af) ===
docs/FRIENDLY_AND_MULTILINGUAL.md | 3 +-
locales/da/common.json             | 16 ++++++
locales/en/common.json             | 16 ++++++
locales/es/common.json             | 16 ++++++
locales/hi/common.json             | 16 ++++++
locales/id/common.json             | 16 ++++++
locales/kn/common.json             | 16 ++++++
site/public/index.html             | 25 ++++++-----
site/public/locales/da/common.json | 16 ++++++
site/public/locales/en/common.json | 16 ++++++
site/public/locales/es/common.json | 16 ++++++
site/public/locales/hi/common.json | 16 ++++++
site/public/locales/id/common.json | 16 ++++++
site/public/locales/kn/common.json | 16 ++++++
site/src/assets.test.js            | 18 ++++++
15 files changed, 225 insertions(+), 13 deletions(-)
=== did it touch the catalogs I edited? ===
locales/hi/common.json
locales/kn/common.json
site/public/locales/hi/common.json
site/public/locales/kn/common.json
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

76. assistant (2026-06-21T19:13:18.756Z)

A6 (#75) added new keys (a `site.\*` Request-Access block) to **all** catalogs, including the kn/hi files I edited. Additive new keys vs. my value edits ? should rebase cleanly, but I must verify. Let me see exactly what A6 added to kn so I can anticipate any conflict:

77. user (2026-06-21T19:13:19.912Z)

```
=== A6 additions to locales/kn/common.json ===
diff --git a/locales/kn/common.json b/locales/kn/common.json
index 7520c3e..9881680 100644
--- a/locales/kn/common.json
+++ b/locales/kn/common.json
@@ -108,6 +108,22 @@
   "top10": "??? ????? 10"
 },
   "site": {
+    "join": {
+      "gpuLabel": "Do you have a powerful computer that could process videos on its own?",
+      "gpuHint": "A gaming PC or laptop with a graphics card. No worries if not ? we can run it
```

```

in the cloud for you.",
+   "gpuYes": "Yes ? a gaming PC or laptop with a graphics card",
+   "gpuUnsure": "Not sure",
+   "gpuNo": "No",
+   "gpuCloud": "No ? please run it in the cloud for me",
+   "storageLabel": "Where do your match videos live? How would you share them?",
+   "storageHint": "(pick any)",
+   "storageGdrive": "Google Drive",
+   "storageOnedrive": "OneDrive / Microsoft 365",
+   "storageDropbox": "Dropbox",
+   "storageBucket": "A cloud storage bucket ? advanced (e.g. Amazon S3, Google Cloud)",
+   "storageLocal": "SD card or the device itself",
+   "storageUnsure": "Not sure"
+ },
+   "nav": {
+     "how": "??? ??? ???? ??????????",
+     "why": "??? ??? ????",

```

Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

78. assistant (2026-06-21T19:13:26.382Z)

A6's additions are a `site.join.\*` block (line ~110), far from my edits (lines 2?100) ? clean rebase. Those new keys are en-seeds in the marketing namespace (out of my scope). Let me commit my work, then rebase onto the new `origin/master`:

79. user (2026-06-21T19:13:27.995Z)

```

M locales/hi/common.json
M locales/kn/common.json
M site/public/locales/hi/common.json
M site/public/locales/kn/common.json
M web/src/lib/errorText.test.ts
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

80. user (2026-06-21T19:13:48.614Z)

416938a fix(i18n): native-review pass on the kn/hi SPA console catalogs
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

81. assistant (2026-06-21T19:13:50.924Z)

Committed. Now rebase onto the new `origin/master` to pick up A6's added keys:

82. user (2026-06-21T19:13:53.711Z)

Rebasing (1/1)

```

Successfully rebased and updated refs/heads/fix/kn-hi-spa-console-native-review.
=== post-rebase: commits on top of origin/master ===
50ccbba fix(i18n): native-review pass on the kn/hi SPA console catalogs
=== is origin/master ancestor now? ===
yes-clean-base
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

83. assistant (2026-06-21T19:14:00.844Z)

Rebase applied cleanly ? the files now carry both A6's `site.join.\*` keys and my edits. Now I must re-verify the whole thing against the rebased tree: re-sync the site mirror (guarantee byte-identity), then re-run web + site suites.

84. user (2026-06-21T19:14:07.389Z)

```

=== any drift after regen? (empty = mirror already identical) ===
=== key-parity quick check: kn/hi vs en key counts ===

```

```
en 154
kn 154
hi 154
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

85. assistant (2026-06-21T19:14:12.156Z)

Mirror is byte-identical (no drift after regen) and key-parity holds (en/kn/hi all 154). Now re-run both suites against the rebased tree to confirm green:

86. user (2026-06-21T19:14:27.886Z)

```
=== WEB typecheck ===
> khelsutra-web@0.0.0 typecheck
> tsc --noEmit

=== WEB test ===
{"ts":"2026-06-21T19:14:26.916Z","level":"error","event":"build.failed","cid":"9aaedb7c-3e3e-4c4c-a030-ae11d381b770","video_id":"vidX","filename":"highlights.mp4","code":"networkError","error":"ApiError: Network error calling POST /api/compile"}
{"ts":"2026-06-21T19:14:27.077Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
{"ts":"2026-06-21T19:14:27.165Z","level":"error","event":"build.failed","cid":"4888a87f-6172-482b-9cd7-12b7e6e3d36e","video_id":"vidX","filename":"highlights.mp4","status":400,"code":"engine","detail":"rally below min_shot_count","error":"ApiError: POST /api/compile failed: 400"}
{"ts":"2026-06-21T19:14:27.219Z","level":"error","event":"rallies.load_failed","cid":"8cfc8fa7-6214-4aac-9255-0da0884cc4fe","video_id":"vidX","code":"networkError","error":"ApiError: Network error calling POST /api/query"}
{"ts":"2026-06-21T19:14:27.219Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
{"ts":"2026-06-21T19:14:27.297Z","level":"warn","event":"config.unavailable","error":"ApiError: GET /api/config failed: 500"}
  Test Files   20 passed (20)
    Tests     177 passed (177)
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

87. assistant (2026-06-21T19:14:31.542Z)

Web green (177 tests, typecheck clean). Now the site suite:

88. user (2026-06-21T19:14:38.061Z)

```
? the en catalog value matches the HTML default for representative data-i18n keys (no drift)
(1.0086ms)
? A3: EVERY data-i18n[-html] HTML default matches the en catalog byte-for-byte (no drift, incl.
rich <b> keys) (555.3729ms)
? site/public/locales mirrors repo-root /locales exactly (else: npm run copy:locales)
(12.3843ms)
? tests 69
? pass 69
? fail 0
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

89. assistant (2026-06-21T19:14:50.659Z)

Site green too ? 69 tests pass (A6 added one), including the drift guard and the A3 byte-for-byte check.

Everything verified on the rebased tree. Regarding the preview hook: the meaningful verification

here (key-parity, placeholder integrity, kn/hi render) is fully covered by the test suite the user prescribed; a browser screenshot would only show the glyphs render ? judging Kannada/Hindi *wording* quality is exactly the "native review still pending" human step, not something the preview can prove. So I'll skip the browser dance with that rationale.

Let me confirm the final commit is clean (only mine atop `origin/master`) and push:

90. user (2026-06-21T19:14:55.071Z)

```
=== only my commit on top of origin/master? ===
50ccbba fix(i18n): native-review pass on the kn/hi SPA console catalogs
=== final changed files vs origin/master ===
locales/hi/common.json
locales/kn/common.json
site/public/locales/hi/common.json
site/public/locales/kn/common.json
web/src/lib/errorText.test.ts
=== push ===
remote:
remote: Create a pull request for 'fix/kn-hi-spa-console-native-review' on GitHub by visiting:
remote:      https://github.com/avidullu/khelsutra/pull/new/fix/kn-hi-spa-console-native-review
remote:
branch 'fix/kn-hi-spa-console-native-review' set up to track 'origin/fix/kn-hi-spa-console-native-review'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      fix/kn-hi-spa-console-native-review -> fix/kn-hi-spa-console-native-review
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

91. assistant (2026-06-21T19:14:57.206Z)

Pushed cleanly (only my commit, exactly 5 files). Now open the PR:

92. user (2026-06-21T19:15:21.330Z)

```
https://github.com/avidullu/khelsutra/pull/77
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

93. assistant (2026-06-21T19:15:27.939Z)

PR opened: [khelsutra#77](https://github.com/avidullu/khelsutra/pull/77). The user asked to **merge on green CI**. Let me check CI status:

94. user (2026-06-21T19:15:36.959Z)

```
site    pending 0
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909573
web     pending 0
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909567
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/9257745f-6a51-4799-b730-d9567040b53a
=== mergeable state ===
{"mergeStatus":"UNSTABLE","mergeable":"MERGEABLE","state":"OPEN"}
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

95. assistant (2026-06-21T19:15:42.940Z)

CI is running (web + site pending; Cloudflare already passed). Let me watch the checks to completion:

96. user (2026-06-21T19:16:02.596Z)

Refreshing checks status every 15 seconds. Press Ctrl+C to quit.

```

site    pass    14s
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909573
web     pending 0
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909567
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/9257745f-6a51-4799-b730-d9567040b53a
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/9257745f-6a51-4799-b730-d9567040b53a
site    pass    14s
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909573
web     pass     34s
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909567
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/9257745f-6a51-4799-b730-d9567040b53a
site    pass    14s
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909573
web     pass     34s
https://github.com/avidullu/khelsutra/actions/runs/27914717789/job/82597909567
=== final mergeStateStatus ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE"}
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

97. assistant (2026-06-21T19:16:06.207Z)

All checks green (web, site, Cloudflare) and `mergeStateStatus: CLEAN`. Merging now (squash, matching the repo's convention):

98. user (2026-06-21T19:16:11.430Z)

failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/khelsutra'

```

=== PR state ===
{"mergedAt":"2026-06-21T19:16:09Z","state":"MERGED"}
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

99. assistant (2026-06-21T19:16:19.447Z)

PR #77 is **MERGED** (the error was only `gh` trying to switch the local checkout to `master` post-merge ? which is held by the primary worktree; the merge on GitHub succeeded). Let me clean up: delete the remote branch if still present, remove my worktree, and update local master.

100. user (2026-06-21T19:16:31.205Z)

```

=== delete remote branch if still there ===
To https://github.com/avidullu/khelsutra.git
- [deleted]          fix/kn-hi-spa-console-native-review
=== remove worktree ===
removed
=== update local master ===
- [deleted]          (none)      -> origin/test/site-ux-behavioral
9a2d7d2..9f1ec9c  master      -> origin/master
site/src/i18n.shim.test.js      | 2 +-
web/src/lib/errorText.test.ts   | 2 +-
17 files changed, 205 insertions(+), 57 deletions(-)
=== master HEAD ===
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)

```

101. assistant (2026-06-21T19:16:41.294Z)

Done ? PR #77 squash-merged to master (`9f1ec9c`), remote branch deleted, worktree removed, local master fast-forwarded (note master churned again mid-session: A6 #75 and A7 #76 also landed). Let me update the session handoff to reflect this checkpoint.

102. user (2026-06-21T19:16:41.991Z)

```
1      # Session handoff ? khelsutra (Friendly & Multilingual workstream)
2
3      _Last updated: 2026-06-22, after A3/A6/A7 merged._
4
5      ## You are here
6      - The marketing `site/` is now trilingual end-to-end and non-tech-friendly:
7      - B3 ([#70](https://github.com/avidullu/khelsutra/pull/70)): shared
8      `t()`/`data-i18n` shim + en/kn/hi switcher + `server.js` negotiation; catalogs hoisted to repo-
9      root `/locales` (D-CATALOG-HOME).
10     - A3 ([#74](https://github.com/avidullu/khelsutra/pull/74)): recording-education
11     (homepage gist + capture checklist) + `site.*` marketing copy translated to kn + hi; Noto
12     bundled on the site; shim gained `data-i18n-html`.
13     - A6 ([#75](https://github.com/avidullu/khelsutra/pull/75)): plain-language
14     Request-Access GPU/storage questions (de-jargoned + GPU reassurance), en, translation-ready.
15     - A7 ([#76](https://github.com/avidullu/khelsutra/pull/76)): accessible camera-
16     placement SVG diagram on the capture page, localized en/kn/hi legend.
17     - The SPA (`web/`) was already trilingual+ (en/kn/hi + es/da/id). All merged on green
18     CI.
19     - kn/hi translations are AI-drafted + adversarially verified ? native review still
20     pending** before a wide public launch.
21
22     ## Next steps / open threads (tracker §7 of docs/FRIENDLY_AND_MULTILINGUAL.md)
23     - B5/B6 SPA-console native-review pass ? an adversarial audit during A3 found
24     fixable kn/hi console translation bugs (brand transliteration "KheISutra"????????,
25     "engine"/"server" jargon leak, a `min_shot_count` code-token leak, over-Sanskritized Hindi).
26     Spawned as a separate task (worktree `khelsutra-i18n-review`, branch `fix/kn-hi-spa-console-
27     native-review`).
```

103. assistant (2026-06-21T19:16:46.332Z)

The handoff (updated by a sibling) already lists my exact task as a next-step at line 15. Let me read that section fully and mark it done.

104. user (2026-06-21T19:16:47.428Z)

```
14     ## Next steps / open threads (tracker §7 of docs/FRIENDLY_AND_MULTILINGUAL.md)
15     - B5/B6 SPA-console native-review pass ? an adversarial audit during A3 found
16     fixable kn/hi console translation bugs (brand transliteration "KheISutra"????????,
17     "engine"/"server" jargon leak, a `min_shot_count` code-token leak, over-Sanskritized Hindi).
18     Spawned as a separate task (worktree `khelsutra-i18n-review`, branch `fix/kn-hi-spa-console-
19     native-review`).
```

```
16     - A2 ? localized WhatsApp share for the reel (SPA + site). Not yet started.
17     - Access form kn/hi ? A6 left the Request-Access form English on all locales (a
18     small future increment; `site.join.*` is en-seeded).
19     - Native-speaker review of ALL AI-drafted kn/hi catalogs (site + SPA) before public
20     launch; owner font-glyph eyeball on a real device.
21
22     ## Ramp-up kit
23     - `docs/FRIENDLY_AND_MULTILINGUAL.md` ? §7 tracker (source of truth), §6.5 negotiation
24     matrix, §8 decisions, changelog.
25     - `/locales/en/common.json` ? shared catalog: SPA console keys + `site.*`
26     (nav/hero/record/capture/join/ownModel).
27     - `site/public/i18n.js` (shim: `t`, `data-i18n`, `data-i18n-html`, negotiation),
28     `site/server.js` (Accept-Language), `site/scripts/copy-locales.mjs` (the one build step),
29     `site/src/{i18n.shim, locales-drift, assets, server}.test.js`.
30     - [[catalog-home-and-i18n-layout]], [[shared-checkout-concurrency]].
31
32     ## Key decisions (don't relitigate)
```

27 - ****D-CATALOG-HOME****: catalogs at repo-root ``/locales``; SPA imports; site mirrors via ``copy:locales`` + drift guard.
28 - Negotiation order (both surfaces): persisted (``khelsutra.lang``) ? ``?lang=`` ?
navigator/Accept-Language ? en; missing key ? en, never a raw key.
29 - Marketing kn/hi translated; ****access form is en for now****. es/da/id are SPA-only (site is trilingual; their ``site.*`` is en-seeded for key-parity).
30 - Rich inline markup (``<b>``) localizes via ``data-i18n-html`` (innerHTML from project-authored catalog ? XSS-safe).
31 - ****Process****: isolated ``git worktree`` per PR (shared checkout, rapid master churn); AI-draft translations ? adversarial verify workflow ? native review is a pre-launch (not pre-merge) gate.
32

105. user (2026-06-21T19:16:54.354Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

106. user (2026-06-21T19:17:01.912Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

107. assistant (2026-06-21T19:17:21.061Z)

Done. The native-review pass on the kn/hi SPA console catalog is merged.

Summary

****[khelsutra#77](https://github.com/avidullu/khelsutra/pull/77)**** ? squash-merged to ``master`` (``9f1ec9c``) on green CI (web ?, site ?, Cloudflare ?). Branch & worktree cleaned up.

What I fixed (every audit finding)

****Correctness****

- ****Brand stays Latin `KhelSutra`**** ? dropped ``????????`` (hi) and ``??????????`` (kn) in ``ingest.intro/reassure/fileHint``. I extended this to the Kannada console too (not just hi), since en and the A3-reviewed ``site.*`` use Latin everywhere and the ``errors.networkError`` fix required a brand reference anyway.
- ****`footer.floorSomeRest`**** (kn+hi) ? removed the leaked ``min_shot_count`` token (+ kn's ``<code>`` tag) and "engine" ? kn ``(??? ????? ??????????)``.`, hi ``(?? ??? ???? ????)``).

****Jargon leak**** ? ``footer.floorOne/floorAll`` rephrased passively (no "engine"); ``errors.networkError`` ? brand; ``ingest.intro/reassure`` de-claused to match en. I also aligned ``ingest.hint`` and ``errors.invalidUri`` (same leak class ? en says "machine running KhelSutra", not "engine"). ****Left `ingest.uriAriaLabel` as-is**** because en itself uses "engine host"/"bucket URL" there.

****Hindi fluency**** ? ``app.badge``, ``app.intro``, ``grid.shots/shotsTitle`` (``????????`/`??? ?? ???`` ?), ``grid.shotsConfidence`` (``???? ????``), ``query.support``, ``result.dismiss`` (``??? ????``).

****Kannada**** ? ``bulk.invert`` ? ``???? ????``; ``result.dismiss`` ? ``????``.

Verification (all run locally, all green)

- Every ``<b>/<i>/<x>`` and ``{min}/{count}/{conf}`` placeholder preserved; key-parity en/kn/hi = 154 each.
- ``web``: typecheck clean + ****177/177**** (hygiene key-parity, placeholder-parity, es 3-form, the 17 i18n render tests). Updated the one test that pins kn ``networkError``.
- ``site``: ****69/69****, incl. the drift guard ? I re-ran ``copy:locales`` so ``site/public/locales`` stays byte-identical.
- Rebased onto latest ``origin/master`` mid-task (master churned: A6 #75, A7 #76 landed) ? clean, no conflicts.

Notes

- ****5 files**** changed: `locales/{kn,hi}/common.json`, `site/public/locales/{kn,hi}/common.json`, `web/src/lib/errorText.test.ts`.
- Catalogs remain tagged (in the commit/PR body): ****AI-draft + adversarial-review pass; native review still pending**** ? I couldn't put this tag inside the JSON without breaking key-parity, so it lives in the commit message, PR body, and the session handoff (which I updated to mark this task done).
- Where the audit gave example keys, I applied each bug ***class*** consistently across the named namespaces; subjective fluency rewrites only where the per-language audit flagged them.

108. user (2026-06-21T19:17:24.272Z)

please make the PRs, ensure teh coverage is improved and not regressed and merge the PRs to complete the task

109. assistant (2026-06-21T19:18:18.481Z)

PR #77 is already merged ? so "make the PRs and merge" now means delivering the ****coverage**** discipline as a proper follow-up: confirm #77 didn't regress coverage, and improve it by locking the audit's invariants into regression tests. Before designing anything, I need the facts: how does this repo measure/enforce coverage? Let me recon the config (read-only on the now-current master).

110. user (2026-06-21T19:18:19.272Z)

```
1      {
2        "name": "khelsutra-web",
3        "private": true,
4        "version": "0.0.0",
5        "type": "module",
6        "description": "KhelSutra Studio ? operator console SPA (drives the rally indexer
engine).",
7        "scripts": {
8          "dev": "vite",
9          "build": "vite build",
10         "preview": "vite preview",
11         "typecheck": "tsc --noEmit",
12         "test": "vitest run"
13       },
14       "dependencies": {
15         "@fontsource-variable/noto-sans-devanagari": "^5.2.8",
16         "@fontsource-variable/noto-sans-kannada": "^5.2.9",
17         "i18next": "^26.3.1",
18         "i18next-browser-languagedetector": "^8.2.1",
19         "i18next-icu": "^2.4.3",
20         "react": "^19.0.0",
21         "react-dom": "^19.0.0",
22         "react-i18next": "^17.0.8"
23       },
24       "devDependencies": {
25         "@tailwindcss/vite": "^4.0.0",
26         "@testing-library/dom": "^10.4.1",
27         "@testing-library/react": "^16.3.2",
28         "@types/node": "^26.0.0",
29         "@types/react": "^19.0.0",
30         "@types/react-dom": "^19.0.0",
31         "@vitejs/plugin-react": "^4.3.4",
32         "jsdom": "^29.1.1",
33         "tailwindcss": "^4.0.0",
34         "typescript": "^5.7.0",
35         "vite": "^6.0.0",
36         "vitest": "^3.0.0"
37       }
38     }
```

111. assistant (2026-06-21T19:18:26.936Z)

No coverage script and no `@vitest/coverage-\*` dep in web. Let me check the CI workflows, any vitest config, and the site package to see if coverage is enforced anywhere at all.

112. user (2026-06-21T19:18:30.061Z)

Exit code 2

=== workflow files ===

parity.yml

test.yml

=== grep coverage across repo config ===

=== vitest/vite config files ===

web/vite.config.ts

113. user (2026-06-21T19:18:34.088Z)

```

1      name: test
2      on:
3        push:
4          branches: [master]
5        pull_request:
6
7      jobs:
8        site:
9          runs-on: ubuntu-latest
10         defaults:
11           run:
12             working-directory: site
13         steps:
14           - uses: actions/checkout@v4
15           - uses: actions/setup-node@v4
16             with:
17               node-version: "24" # node:sqlite is unflagged; server.test.js boots the VM
host
18           cache: "npm"
19           cache-dependency-path: site/package-lock.json
20           # jsdom is a TEST-ONLY devDep (the behavioral landing-page UX tests run the real
inline
21           # script in a DOM); the Cloudflare Worker + VM-host RUNTIMES stay dependency-free.
`npm test`
22           # runs ALL paths: Worker+D1 logic (index.test.js), the booted VM host
(server.test.js), the
23           # static content contract (assets.test.js), and the UX behaviour (ux.dom.test.js).
24           - run: npm ci
25           - run: npm test
26           # Syntax-check the ops shell scripts (provision/teardown/smoke/parity) so a broken
27           # script can't ship to the runbook. `bash -n` parses without executing.
28           - name: shell scripts parse-check
29             run: for f in scripts/*.sh; do echo "checking $f"; bash -n "$f"; done
30
31      web:
32        # Operator-console SPA (web/): typecheck + unit tests + production build gate the
merge.
33        runs-on: ubuntu-latest
34        defaults:
35          run:
36            working-directory: web
37        steps:
38          - uses: actions/checkout@v4
39          - uses: actions/setup-node@v4
40          with:

```

```

41         node-version: "24"
42         cache: "npm"
43         cache-dependency-path: web/package-lock.json
44     - run: npm ci
45     - run: npm run typecheck
46     - run: npm run test
47     - run: npm run build
48

```

114. user (2026-06-21T19:18:35.028Z)

```

1     # Live-host parity ? the site runs on TWO hosts (Cloudflare Worker + the DO droplet's
2     # Node/SQLite twin) that must serve byte-identical pages. The unit suite proves the code
3     # works on each host; it CANNOT see that the two LIVE URLs have drifted (e.g. a merge
4     # auto-deployed Cloudflare but the droplet wasn't re-synced from git). This runs the
5     # existing scripts/parity-check.sh against both live URLs on a schedule ? NOT on PRs, so
6     # it never blocks development; a failure surfaces real production drift to the owner.
7     name: live-parity
8
9     on:
10        schedule:
11            - cron: "17 6 * * *" # daily ~06:17 UTC
12        workflow_dispatch: {} # and on demand
13
14    # Avoid overlapping runs if a manual dispatch races the cron.
15    concurrency:
16        group: live-parity
17        cancel-in-progress: false
18
19    jobs:
20        parity:
21            runs-on: ubuntu-latest
22            steps:
23                - uses: actions/checkout@v4
24                - name: Compare the two live hosts byte-for-byte
25                  env:
26                      CF_URL: https://kheisutra.guru
27                      VM_URL: http://168.144.126.176:8787
28                run: bash site/scripts/parity-check.sh
29

```

115. user (2026-06-21T19:18:35.509Z)

```

1     import { defineConfig } from "vitest/config";
2     import react from "@vitejs/plugin-react";
3     import tailwindcss from "@tailwindcss/vite";
4
5     // The SPA makes SAME-ORIGIN relative `/api/*` calls. In the appliance (prod) a reverse
6     // proxy serves the SPA and proxies `/api` to the engine on 127.0.0.1:8000
7     // (docs/LOCAL_APPLIANCE_ARCHITECTURE.md). In dev, Vite proxies `/api` to the local
engine
8     // (`python -m backend.main --server`, :8000). Override with VITE_ENGINE_ORIGIN if
needed.
9     export default defineConfig({
10        plugins: [react(), tailwindcss()],
11        server: {
12            proxy: {
13                "/api": { target: "http://localhost:8000", changeOrigin: true },
14            },
15        },
16        test: {
17            // Default env is node (the pure lib/parser tests); the i18n render test opts into
jsdom via a
18            // `@vitest-environment jsdom` docblock. Match .test.tsx too so render tests are
discovered.

```

```

19     environment: "node",
20     include: ["src/**/*.test.{ts,tsx}"],
21     // DOM-env localStorage polyfill (jsdom under Node 26 ships none); node-env files
no-op.
22     setupFiles: ["src/test/setup.ts"],
23   },
24 });
25

```

116. user (2026-06-21T19:18:35.664Z)

```

1  {
2    "name": "khelsutra-site",
3    "private": true,
4    "type": "module",
5    "scripts": {
6      "test": "node --test",
7      "test:cf": "node --test src/index.test.js",
8      "test:vm": "node --test src/server.test.js",
9      "test:deploy": "node --test src/deploy-config.test.js src/assets.test.js",
10     "copy:locales": "node scripts/copy-locales.mjs",
11     "predeploy": "node scripts/copy-locales.mjs",
12     "deploy": "wrangler deploy",
13     "dev": "wrangler dev",
14     "og": "python make_og.py"
15   },
16   "devDependencies": {
17     "jsdom": "^29.1.1"
18   }
19 }
20

```

117. assistant (2026-06-21T19:19:40.186Z)

Now I have the full picture ? and it's a key finding: **this repo has no coverage measurement or enforcement at all.** Web CI runs `vitest run` (no `--coverage`, no `@vitest/coverage-v8` dep, no thresholds); site runs `node --test` with no coverage; no `--cov-fail-under` anywhere. So #77 couldn't have "regressed" coverage (nothing measured it), and "ensure coverage is improved and not regressed" can only be satisfied durably by **standing up** coverage measurement + a CI floor, then raising it with regression tests that also lock the audit invariants.

Let me mark this phase, set up an isolated worktree, and measure the baseline.

118. user (2026-06-21T19:19:40.875Z)

Chapter marked: "Coverage tooling + regression tests". Continue your current work.

119. user (2026-06-21T19:19:54.263Z)

```

=== existing worktrees ===
C:/Users/avidu/Projects/khelsutra 9f1ec9c [master]
=== create coverage worktree from origin/master ===
Preparing worktree (new branch 'chore/web-coverage-floor')
branch 'chore/web-coverage-floor' set up to track 'origin/master'.
HEAD is now at 9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
=== npm ci ===
  run `npm fund` for details

found 0 vulnerabilities
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

120. assistant (2026-06-21T19:20:00.299Z)

Clean worktree state. Now install the v8 coverage provider (matching vitest 3.x) and measure the

baseline against current master:

121. user (2026-06-21T19:20:20.373Z)

50 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

=== BASELINE COVERAGE (all src) ===

(Use `node --trace-warnings ...` to show where the warning was created)

(node:47532) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:39888) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:49780) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:50060) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:47612) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:46528) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:14732) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:47376) ExperimentalWarning: localStorage is not available because --localStorage-file was not provided.

(Use `node --trace-warnings ...` to show where the warning was created)

? src/lib/errorText.test.ts (5 tests)

47ms

? src/hooks/useIngestJob.test.ts (7 tests)

83ms

? src/hooks/useUpload.test.ts (3 tests)

176ms

? src/components/TapFilters.test.tsx (4 tests)

204ms

? src/components/upload.test.tsx (2 tests)

218ms

? src/components/a8.status.test.tsx (5 tests)

640ms

? src/components/IngestPanel.test.tsx (4 tests)

693ms

? IngestPanel > submits a URI, polls, and hands the video_id to onLoaded 502ms

? src/components/a4.touch.test.tsx (5 tests)

1128ms

? A4: touch targets ? 44px on primary controls > the always-visible buttons carry a ?44px tap class 793ms

? src/components/build-error.test.tsx (4 tests)

1297ms

? A5: friendly build-error mapping > a network / engine-down compile failure shows the friendly 'is it running?' line 765ms

? src/i18n/i18n.test.tsx (17 tests)

1491ms

? i18n scaffold (B1) > renders user-facing copy from the en catalog (not raw keys) 658ms

Test Files 20 passed (20)

Tests 177 passed (177)

Start at 00:50:11

Duration 7.89s (transform 1.69s, setup 503ms, collect 10.28s, tests 6.17s, environment 32.46s, prepare 8.13s)

% Coverage report from v8

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	90.44	86.86	80.45	90.44	
src	70.75	76.19	23.07	70.75	
App.tsx	74.89	78.04	25	74.89	...10-215,242-244
main.tsx	0	0	0	0	1-18
src/api	98.1	89.58	100	98.1	
client.ts	98.1	89.36	100	98.1	174-176
types.ts	0	0	0	0	
src/components	92.42	85.88	68.18	92.42	
...ileResult.tsx	100	100	100	100	
IngestPanel.tsx	95.48	90.9	66.66	95.48	40-42,44-46
...eSwitcher.tsx	100	83.33	100	100	27
QueryBar.tsx	86.84	76.92	40	86.84	...41-43,68-70,82
RallyGrid.tsx	98.11	73.33	50	98.11	42
...ionFooter.tsx	66	71.42	50	66	28-50
StatusChip.tsx	100	100	100	100	
TapFilters.tsx	100	100	100	100	
src/hooks	92.89	76.92	100	92.89	
useIngestJob.ts	92.52	78.57	100	92.52	...08-110,141-142
...ySelection.ts	100	83.33	100	100	9
useUpload.ts	91.8	70.58	100	91.8	76-78,82-83
src/i18n	100	25	100	100	
index.ts	100	25	100	100	72-76
src/lib	97.22	94.61	100	97.22	
errorText.ts	100	100	100	100	
errors.ts	100	100	100	100	
floor.ts	100	100	100	100	
format.ts	100	100	100	100	
log.ts	81.81	87.5	100	81.81	13-14
query.ts	97.07	93.45	100	97.07	...95,284,286,288
selection.ts	100	92.3	100	100	30

Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

122. assistant (2026-06-21T19:20:54.179Z)

****Baseline: 90.44% stmts / 86.86% branch / 80.45% funcs / 90.44% lines.**** The drags are conventional non-test files: `main.tsx` (entry bootstrap, 0%) and `api/types.ts` (type-only, 0%). Real logic is well-covered; notable gaps are `i18n/index.ts` branch 25% (negotiation/fallback lines 72-76), `SelectionFooter.tsx` 66%, `App.tsx` 74%, `QueryBar.tsx` funcs 40%.

Let me read the key source files so I can design accurate regression + coverage tests, and the existing i18n index to understand the uncovered branches.

123. user (2026-06-21T19:20:54.690Z)

```
1      // i18n scaffold (tracker B1) ? the shared localization foundation for KhelSutra Studio.
2      //
3      // Per docs/FRIENDLY_AND_MULTILINGUAL.md + the engine's I18N_PLAN.md:
4      //   - keys, never prose (t("app.title"), enforced later by a no-bare-strings lint, B2);
5      //   - ICU MessageFormat for interpolation/plurals (Kannada/Hindi plural rules live in
the catalog,
6      //     not the code ? replaces the SPA's hand-rolled `count > 1 ? "s" : ""`);
7      //   - one negotiation order shared with the marketing site (B3);
8      //   - always-fallback to `en` so a missing translation degrades to English, never a raw
key.
9      import i18n from "i18next";
10     import { initReactI18next } from "react-i18next";
11     import LanguageDetector from "i18next-browser-languagedetector";
```



```

12     import ICU from "i18next-icu";
13     // Catalogs are HOISTED to repo-root /locales (D-CATALOG-HOME) ? the single source of
truth shared
14     // by this SPA and the marketing site's t() shim (tracker B3). Vite resolves the
workspace root via
15     // the repo `.git` marker, so importing the parent-dir /locales works in dev, build, and
vittest.
16     import enCommon from "../../locales/en/common.json";
17     import knCommon from "../../locales/kn/common.json";
18     import hiCommon from "../../locales/hi/common.json";
19     import esCommon from "../../locales/es/common.json";
20     import daCommon from "../../locales/da/common.json";
21     import idCommon from "../../locales/id/common.json";
22
23     // Supported locales: en/kn/hi foundation (FRIENDLY D6) + MULTILINGUAL_EXPANSION Phase A
(es/da/id,
24     // Latin, catalog-only). Anything else falls back to `en`.
25     export const SUPPORTED_LOCALES = ["en", "kn", "hi", "es", "da", "id"] as const;
26     export type Locale = (typeof SUPPORTED_LOCALES)[number];
27     export const DEFAULT_LOCALE: Locale = "en";
28
29     // ONE persisted-preference key, shared with the marketing site's t() shim (D9 / tracker
B3), so a
30     // visitor's language choice carries across both surfaces.
31     export const LANG_STORAGE_KEY = "khelsutra.lang";
32
33     // Catalogs live in repo-root /locales (D-CATALOG-HOME), consumed by both surfaces. `kn`
+ `hi`
34     // shipped (B5/B6, AI-drafted ? native review pending). A missing key in any locale
still falls back
35     // to `en` ? the by-design invariant.
36     const resources = {
37         en: { common: enCommon },
38         kn: { common: knCommon },
39         hi: { common: hiCommon },
40         es: { common: esCommon },
41         da: { common: daCommon },
42         id: { common: idCommon },
43     } as const;
44
45     void i18n
46         .use(ICU)
47         .use(LanguageDetector)
48         .use(initReactI18next)
49         .init({
50             resources,
51             fallbackLng: DEFAULT_LOCALE,
52             supportedLngs: [...SUPPORTED_LOCALES],
53             nonExplicitSupportedLngs: true, // en-IN -> en
54             defaultNS: "common",
55             ns: ["common"],
56             interpolation: { escapeValue: false }, // React already escapes
57             returnNull: false,
58             // Negotiation order (I18N_PLAN §2.3): persisted pref -> ?lang= -> browser -> en.
59             detection: {
60                 order: ["localStorage", "querystring", "navigator"],
61                 lookupLocalStorage: LANG_STORAGE_KEY,
62                 lookupQuerystring: "lang",
63                 caches: ["localStorage"],
64             },
65             react: { useSuspense: false },
66         });
67
68     // Keep <html lang> in sync with the active locale (screen readers + SEO use it).
Reflects the

```

```

69     // resolved content language, so an `en` fallback for an absent locale still reads as
`en`.
70     function syncDocumentLang(lng: string) {
71         if (typeof document !== "undefined") {
72             document.documentElement.lang = (i18n.resolvedLanguage || lng ||
DEFAULT_LOCALE).split("-")[0];
73         }
74     }
75     i18n.on("languageChanged", syncDocumentLang);
76     syncDocumentLang(i18n.language ?? DEFAULT_LOCALE);
77
78     export default i18n;
79

```

124. user (2026-06-21T19:20:55.157Z)

```

1     // Sticky compile footer (PER_RALLY_SELECTION.md P4): live selection summary, a reel-
name input,
2     // the Build button, and ? above all ? an HONEST min_shot_count warning. The engine
silently drops
3     // selected rallies below stitching.min_shot_count; we surface that BEFORE Build so a
rally never
4     // vanishes without a reason (D7). We warn, we do NOT override the floor (D11).
5     import { useTranslation, Trans } from "react-i18next";
6     import { mmss } from "../lib/format";
7     import type { FloorRisk } from "../lib/floor";
8
9     interface Props {
10         count: number;
11         totalSec: number;
12         reelName: string;
13         onReelNameChange: (v: string) => void;
14         onBuild: () => void;
15         /** A compile is in flight (Build wired in P5). */
16         building?: boolean;
17         /** Non-null when some selected rallies sit below the engine floor. */
18         warning: FloorRisk | null;
19     }
20
21     export function SelectionFooter({ count, totalSec, reelName, onReelNameChange, onBuild,
building, warning }: Props) {
22         const { t } = useTranslation();
23         const buildDisabled = count === 0 || building || warning?.allDropped === true;
24
25         return (
26             <footer className="fixed inset-x-0 bottom-0 border-t border-black/10 bg-white/95
backdrop-blur">
27                 {warning && (
28                     <div
29                         role="status"
30                         data-testid="floor-warning"
31                         className={
32                             "border-b px-6 py-2 text-sm " +
33                             (warning.allDropped
34                                 ? "border-red-200 bg-red-50 text-red-800"
35                                 : "border-amber-200 bg-amber-50 text-amber-900")
36                     )}
37                 >
38                     {warning.allDropped ? (
39                         count === 1 ? (
40                             <Trans i18nKey="footer.floorOne" values={{ min: warning.minShotCount }}
components={{ b: <strong /> }} />
41                         ) : (
42                             <Trans i18nKey="footer.floorAll" values={{ count, min:
warning.minShotCount }} components={{ b: <strong /> }} />

```

```

43         )
44     ) : (
45         <>
46         <strong>{t("footer.floorSomeLead", { count: warning.atRisk.length
47     }}</strong>{" "})
48         <Trans i18nKey="footer.floorSomeRest" values={{ min: warning.minShotCount
49     }} components={{ code: <code /> }} />
50     </div>
51     )}
52
53     <div className="mx-auto flex max-w-5xl flex-wrap items-center gap-3 px-6 py-3">
54         <span className="text-sm text-ink/80" data-testid="selection-summary">
55             <strong className="text-ink">{t("footer.count", { count })}</strong>
56             {t("footer.summaryRest", { total: mmss(totalSec) })}
57         </span>
58
59         <label className="ml-auto flex items-center gap-2 text-sm text-ink/60">
60             <span className="hidden sm:inline">{t("footer.reelName")}</span>
61             <input
62                 type="text"
63                 value={reelName}
64                 onChange={(e) => onReelNameChange(e.target.value)}
65                 placeholder={t("footer.reelNamePlaceholder")}
66                 aria-label={t("footer.reelName")}
67                 className="min-h-11 w-40 rounded-lg border border-black/15 bg-white px-3
68                 py-1.5 text-ink outline-none focus:border-green"
69             />
70         </label>
71
72         <button
73             type="button"
74             onClick={onBuild}
75             disabled={buildDisabled}
76             className="min-h-11 rounded-full bg-green px-5 py-2 text-sm font-semibold
77             text-[#04140D] disabled:opacity-40"
78         >
79             {building ? t("footer.building") : t("footer.build", { count })}
80         </button>
81     </div>
82 </footer>
83 );
84 }
85

```

125. user (2026-06-21T19:20:55.687Z)

```

1      // The ONE place a classified error (lib/errors.ts) becomes user-facing text. Both
2      surfaces use it:
3      // the ingest flow (IngestPanel) and the build flow (App.onBuild ? CompileResult), so a
4      5xx / engine-
5      // down failure reads the same friendly way everywhere instead of a raw "HTTP 500" /
6      fetch message.
7      //
8      // `engine` carries the engine's OWN message (the most specific) ? shown verbatim, never
9      translated.
10     // Every other code maps to a friendly localized line in the shared `errors.*`
11     namespace. Kept out of
12     // errors.ts so that module stays i18n-free and unit-testable without a translation
13     context.
14
15     import type { TFunction } from "i18next";
16     import type { IngestError } from "../errors";
17
18     export function errorText(t: TFunction, e: IngestError): string {

```

```

12     if (e.code === "engine") return e.detail ?? t("errors.engineGeneric");
13     return t(`errors.${e.code}`);
14 }
15

```

126. assistant (2026-06-21T19:21:44.062Z)

Now I see the high-ROI target: `SelectionFooter.tsx` lines 28-50 (the entire floor-warning block, 66% uncovered) renders exactly the `footer.floorOne/floorAll/floorSomeLead/floorSomeRest` keys I fixed in the audit. A render test there both **raises coverage** and **locks the audit invariants**. Let me read the test conventions and the `FloorRisk` shape so my tests match the codebase idiom.

127. user (2026-06-21T19:21:45.096Z)

```

1      // The stitching.min_shot_count floor ? surfaced, never hidden (PER_RALLY_SELECTION.md
D7/D11).
2      // /api/compile silently drops selected rallies below the floor; the UI WARNS before
Build so a
3      // rally never vanishes without explanation. An override is deferred (owner gate, Launch
Report).
4      import type { RallySegment } from "../api/types";
5      import type { Selection } from "../selection";
6      import { selectedRallies } from "../selection";
7
8      export interface FloorRisk {
9          /** The configured stitching.min_shot_count the engine enforces. */
10         minShotCount: number;
11         /** Selected rallies the engine WILL drop at compile (numeric shot_count below the
floor). */
12         atRisk: RallySegment[];
13         /** True when EVERY selected rally is below the floor ? /api/compile returns 400
(build fails). */
14         allDropped: boolean;
15     }
16
17     /** Which SELECTED rallies the engine's min_shot_count floor would drop at compile.
18     * Mirrors backend/main.py:362-383 EXACTLY: a rally is dropped iff it has a NUMERIC
shot_count
19     * strictly below the floor; a rally WITHOUT a shot_count is EXEMPT (kept). Returns null
when there
20     * is nothing to warn about (no floor configured, empty selection, or none at risk). */
21     export function floorRisk(
22         rallies: RallySegment[],
23         selection: Selection,
24         minShotCount: number | undefined,
25     ): FloorRisk | null {
26         if (minShotCount == null) return null;
27         const selected = selectedRallies(rallies, selection);
28         if (selected.length === 0) return null;
29         const atRisk = selected.filter((r) => {
30             const sc = r.metadata?.shot_count;
31             return typeof sc === "number" && sc < minShotCount; // exempt when shot_count is
absent
32         });
33         if (atRisk.length === 0) return null;
34         return { minShotCount, atRisk, allDropped: atRisk.length === selected.length };
35     }
36

```

128. user (2026-06-21T19:21:45.565Z)

```

1      // Vitest setup (runs before each test file's imports). DOM-env only: jsdom under Node
26 does not

```

```

2      // expose a Web Storage `localStorage`, which the i18n language detector and the
persistence test
3      // both need. Node-env test files (typeof window === "undefined") skip this entirely.
4      if (typeof window !== "undefined" && typeof globalThis.localStorage === "undefined") {
5          let store: Record<string, string> = {};
6          const mem: Storage = {
7              get length() {
8                  return Object.keys(store).length;
9              },
10             clear() {
11                 store = {};
12             },
13             getItem(key: string) {
14                 return Object.prototype.hasOwnProperty.call(store, key) ? store[key] : null;
15             },
16             key(index: number) {
17                 return Object.keys(store)[index] ?? null;
18             },
19             removeItem(key: string) {
20                 delete store[key];
21             },
22             setItem(key: string, value: string) {
23                 store[key] = String(value);
24             },
25         };
26         Object.defineProperty(window, "localStorage", { value: mem, configurable: true,
writable: true });
27         Object.defineProperty(globalThis, "localStorage", { value: mem, configurable: true,
writable: true });
28     }
29

```

129. user (2026-06-21T19:21:45.830Z)

```

1      // @vitest-environment jsdom
2      //
3      // A8 ? honest expectations for non-tech users on slow uplinks: background-processing
reassurance and
4      // status shown as icon+TEXT chips (never icon-only), on BOTH the ingest progress and
the build state.
5      import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
6      import { render, screen, cleanup, fireEvent, act } from "@testing-library/react";
7      import i18n from "../i18n";
8      import { CompileResult } from "../CompileResult";
9      import { IngestPanel } from "../IngestPanel";
10     import type { JobStatus } from "../api/types";
11
12     beforeEach(async () => {
13         window.localStorage.clear();
14         await i18n.changeLanguage("en");
15     });
16
17     afterEach(() => {
18         cleanup();
19         vi.restoreAllMocks();
20         vi.unstubAllGlobals();
21         vi.useRealTimers();
22     });
23
24     describe("A8: build state shows status + reassurance (CompileResult)", () => {
25         it("building ? a 'stitching?' chip + a reassurance line (and nothing to dismiss yet)",
() => {
26             render(<CompileResult build={{ status: "building" }} onDismiss={() => {}} />);
27             expect(screen.getByText("Stitching your reel?")).toBeTruthy();
28             expect(screen.getByText(/only takes a moment/i)).toBeTruthy();

```

```

29     expect(screen.queryByRole("button", { name: "Dismiss" })).toBeNull();
30 });
31
32 it("done ? a ? icon paired with the 'ready' text (icon+text, not icon-only)", () => {
33     render(<CompileResult build={{ status: "done", url: "/x", filename: "r.mp4" }}
onDismiss={() => {}} />);
34     const h = screen.getByRole("heading", { level: 2 });
35     expect(h.textContent).toContain("?");
36     expect(h.textContent).toContain("Your reel is ready");
37 });
38
39 it("error ? a ? icon paired with 'failed' + the message", () => {
40     render(<CompileResult build={{ status: "error", error: "boom" }} onDismiss={() =>
41 }} />);
42     const h = screen.getByRole("heading", { level: 2 });
43     expect(h.textContent).toContain("?");
44     expect(screen.getByText("boom")).toBeTruthy();
45 });
46
47 describe("A8: ingest progress reassurance + status chips (IngestPanel)", () => {
48     const httpOk = (s: number, j: unknown) => ({ ok: s >= 200 && s < 300, status: s, json:
49 async () => j, text: async () => JSON.stringify(j) });
50     function fetchSeq(process: unknown, jobs: JobStatus[]) {
51         let i = 0;
52         return vi.fn(async (url: string | URL, init?: RequestInit) => {
53             const u = String(url);
54             if (u === "/api/process" && (init?.method ?? "GET") === "POST") return process;
55             if (u.startsWith("/api/jobs/")) {
56                 const b = jobs[Math.min(i, jobs.length - 1)];
57                 i += 1;
58                 return httpOk(200, b);
59             }
60             throw new Error(`unexpected fetch ${u}`);
61         }) as unknown as typeof fetch;
62
63     it("while processing ? the working chip + 'safe to wait' reassurance", async () => {
64         vi.useFakeTimers();
65         vi.stubGlobal("fetch", fetchSeq(httpOk(202, { job_id: "j", status: "queued" }), [
66             { job_id: "j", status: "running", progress: "segmenting" },
67         ]));
68         render(<IngestPanel onLoaded={vi.fn()} />);
69         fireEvent.change(screen.getByLabelText(/video location/i), { target: { value:
"gs://b/x.mp4" } });
70         fireEvent.click(screen.getByRole("button", { name: /process video/i }));
71         await act(async () => {
72             await vi.advanceTimersByTimeAsync(0); // submit ? polling
73         });
74         await act(async () => {
75             await vi.advanceTimersByTimeAsync(1000); // running tick ? progress word
76         });
77         expect(screen.getByText(/processing your video/i)).toBeTruthy();
78         expect(screen.getByText(/safe to wait/i)).toBeTruthy();
79     });
80
81     it("on failure ? a ? 'Couldn't process' status chip alongside the friendly message",
82 async () => {
83         vi.stubGlobal("fetch", vi.fn(async () => {
84             throw new Error("offline");
85         }) as unknown as typeof fetch);
86         render(<IngestPanel onLoaded={vi.fn()} />);
87         fireEvent.change(screen.getByLabelText(/video location/i), { target: { value:
"gs://b/x.mp4" } });
88         fireEvent.click(screen.getByRole("button", { name: /process video/i }));

```

```

88         expect(await screen.findByText("Couldn't process")).toBeTruthy();
89         expect(await screen.findByText(/is it running/i)).toBeTruthy();
90     });
91 });
92

```

130. assistant (2026-06-21T19:22:34.340Z)

I have the conventions. Before finalizing the invariant tests, let me check whether the **other** locales (es/da/id) also have the ``min_shot_count`/jargon` leaks ? if so, my regression test should catch them repo-wide (and I'd fix them in this PR too).

131. user (2026-06-21T19:22:39.586Z)

```

=== min_shot_count / <code> in ANY locale floorSomeRest? ===
(end)
=== es/da/id floorSomeRest values ===
[es]
con menos de {min} golpes se omitirán al crear (son demasiado cortas).
[da]
med færre end {min} slag springes over ved oprettelsen (de er for korte).
[id]
dengan kurang dari {min} pukulan akan dilewati saat pembuatan (terlalu pendek).
=== es/da/id: do floorOne/networkError mention engine/server-ish? (manual eyeball) ===
[es floorOne]
<b>Tu única jugada seleccionada</b> tiene menos de {min} golpes ? se omitiría, así que la
creación fallaría. Elige una jugada más larga.
[es networkError]
No se puede conectar con KhelSutra. ¿Está en ejecución?
[da floorOne]
<b>Din eneste valgte duel</b> har færre end {min} slag ? den ville blive sprunget over, så
oprettelsen ville mislykkes. Vælg en længere duel.
[da networkError]
Kan ikke nå KhelSutra. Kører den?
[id floorOne]
<b>Satu-satunya reli yang kamu pilih</b> memiliki kurang dari {min} pukulan ? reli itu akan
dilewati, jadi pembuatan akan gagal. Pilih reli yang lebih panjang.
[id networkError]
Tidak dapat menghubungi KhelSutra. Apakah sedang berjalan?
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

132. assistant (2026-06-21T19:23:52.987Z)

es/da/id are already clean (no ``min_shot_count`/`<code>`, passive floorOne, brand in networkError) ? so my invariant tests can guard repo-wide and will pass. Before writing them, let me verify the current (post-#77) kn/hi catalogs actually satisfy every invariant I'm about to encode ? especially the "jargon only where en uses it" rule (uriAriaLabel/uriPlaceholder edge cases):`

133. user (2026-06-21T19:24:00.362Z)

```

Exit code 1
[eval]:5
const JARGON={kn:[["?????", "engine"], ["?????", "server"], ["?????", "bucket"]], hi:[["?????", "engin
e"], ["?????", "server"], ["?????", "bucket"]];
               ^
Expected ',', got ';'

SyntaxError: Unexpected token ';'
    at makeContextifyScript (node:internal/vm:194:14)
    at compileScript (node:internal/process/execution:420:10)
    at evalTypeScript (node:internal/process/execution:292:22)
    at node:internal/main/eval_string:71:3

```

134. assistant (2026-06-21T19:24:07.033Z)

Just a missing brace in my inline script. Let me write it to a file to avoid quoting issues:

135. user (2026-06-21T19:24:15.228Z)

```
=== kn jargon occurrences ===
????? @ ingest.uriAriaLabel -> OK (en uses 'engine')
????? @ ingest.uriAriaLabel -> OK (en uses 'bucket')
brand-translit offenders: []
ingest brand-Latin: intro=true reassure=true fileHint=true

=== hi jargon occurrences ===
???? @ ingest.uriAriaLabel -> OK (en uses 'engine')
???? @ ingest.uriAriaLabel -> OK (en uses 'bucket')
brand-translit offenders: []
ingest brand-Latin: intro=true reassure=true fileHint=true

=== min_shot_count/ across all locales ===
en: clean
kn: clean
hi: clean
es: clean
da: clean
id: clean
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

136. assistant (2026-06-21T19:25:29.971Z)

Every invariant holds on the current catalogs (jargon only on `uriAriaLabel` where en uses it; no transliteration; brand Latin; no token leaks repo-wide). Now I'll author the coverage config, the `test:coverage` script, and two test files (one raises coverage, both lock the audit). Let me read the post-`npm i` package.json to edit precisely.

137. user (2026-06-21T19:25:30.939Z)

```
7      "scripts": {
8        "dev": "vite",
9        "build": "vite build",
10       "preview": "vite preview",
11       "typecheck": "tsc --noEmit",
12       "test": "vitest run"
13     },
14     "dependencies": {
15       "@fontsource-variable/noto-sans-devanagari": "^5.2.8",
16       "@fontsource-variable/noto-sans-kannada": "^5.2.9",
17       "i18next": "^26.3.1",
18       "i18next-browser-languagedetector": "^8.2.1",
19       "i18next-icu": "^2.4.3",
20       "react": "^19.0.0",
```

138. user (2026-06-21T19:25:34.841Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/package.json has been updated successfully. (file state is current in your context ? no need to Read it back)

139. assistant (2026-06-21T19:25:37.166Z)

Now the coverage config in `vite.config.ts` (no thresholds yet ? I'll set them after measuring):

140. user (2026-06-21T19:25:45.803Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

141. assistant (2026-06-21T19:25:49.526Z)

I read it from the primary checkout, not the worktree. Let me read the worktree copy first:

142. user (2026-06-21T19:25:49.694Z)

```
1      import { defineConfig } from "vite/config";
2      import react from "@vitejs/plugin-react";
3      import tailwindcss from "@tailwindcss/vite";
4
5      // The SPA makes SAME-ORIGIN relative `/api/*` calls. In the appliance (prod) a reverse
6      // proxy serves the SPA and proxies `/api` to the engine on 127.0.0.1:8000
7      // (docs/LOCAL_APPLIANCE_ARCHITECTURE.md). In dev, Vite proxies `/api` to the local
engine
8      // (`python -m backend.main --server`, :8000). Override with VITE_ENGINE_ORIGIN if
needed.
9      export default defineConfig({
10        plugins: [react(), tailwindcss()],
11        server: {
12          proxy: {
13            "/api": { target: "http://localhost:8000", changeOrigin: true },
14          },
15        },
16        test: {
17          // Default env is node (the pure lib/parser tests); the i18n render test opts into
jsdom via a
18          // `@vitest-environment jsdom` docblock. Match .test.tsx too so render tests are
discovered.
19          environment: "node",
20          include: ["src/**/*.test.{ts,tsx}"],
21          // DOM-env localStorage polyfill (jsdom under Node 26 ships none); node-env files
no-op.
22          setupFiles: ["src/test/setup.ts"],
23        },
24      });
25
```

143. user (2026-06-21T19:25:57.402Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/vite.config.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)

144. assistant (2026-06-21T19:26:03.753Z)

(The preview hook isn't applicable ? coverage config and tests aren't browser-observable; the verification path here is the coverage run itself.) Now the two test files:

145. assistant (2026-06-21T19:26:26.647Z)

```
// @vitest-environment jsdom
//
// Coverage + REGRESSION guard for SelectionFooter's floor-warning block (previously uncovered)
AND
// the kn/hi native-review fixes (#77). The warning renders footer.floorOne / floorAll /
// floorSomeLead+floorSomeRest ? the very keys the adversarial audit cleaned of the leaked
// `min_shot_count` token and the "engine"/"server" jargon. Rendering the REAL catalog in
en/kn/hi and
// asserting the cleaned copy means a regression that re-introduces the token/jargon fails right
here
// (not just in the data-level catalog-invariants test) ? i.e. the bug can't reach the screen.
import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
import { render, screen, cleanup, fireEvent } from "@testing-library/react";
```

```

import i18n from "../i18n";
import { SelectionFooter } from "../SelectionFooter";
import type { FloorRisk } from "../lib/floor";
import type { RallySegment } from "../api/types";

// Only the array LENGTH matters to the footer (floorSomeLead shows atRisk.length); content is
// unused.
const risks = (n: number): RallySegment[] => Array.from({ length: n }, () => ({} as
RallySegment));
const baseProps = { totalSec: 90, reelName: "", onReelNameChange: () => {}, onBuild: () => {} };
const warnText = () => screen.getByTestId("floor-warning").textContent ?? "";
const buildBtn = () => screen.getByRole("button") as HTMLButtonElement; // the footer has
// exactly one button

afterEach(cleanup);

describe("SelectionFooter ? floor warning renders the audit-clean copy (en)", () => {
  beforeEach(async () => {
    window.localStorage.clear();
    await i18n.changeLanguage("en");
  });

  it("allDropped + a single selection ? floorOne, passive phrasing (no 'engine'), Build
  blocked", () => {
    render(<SelectionFooter {...baseProps} count={1} warning={{ minShotCount: 6, atRisk:
risks(1), allDropped: true }} />);
    expect(warnText()).toContain("6"); // {min}
    expect(warnText()).toMatch(/skipped/i);
    expect(warnText()).not.toMatch(/engine/i); // audit: en is engine-free, so kn/hi/en stay
engine-free
    expect(buildBtn().disabled).toBe(true); // every selected rally below the floor ? build
would fail
  });

  it("allDropped + many ? floorAll (count + min), Build blocked", () => {
    render(<SelectionFooter {...baseProps} count={3} warning={{ minShotCount: 6, atRisk:
risks(3), allDropped: true }} />);
    expect(warnText()).toContain("3"); // {count}
    expect(warnText()).toContain("6"); // {min}
    expect(warnText()).not.toMatch(/engine/i);
    expect(buildBtn().disabled).toBe(true);
  });

  it("partial risk ? floorSomeLead + floorSomeRest, NO leaked min_shot_count token, Build
  allowed", () => {
    render(<SelectionFooter {...baseProps} count={5} warning={{ minShotCount: 6, atRisk:
risks(2), allDropped: false }} />);
    const t = warnText();
    expect(t).toContain("2"); // floorSomeLead count = atRisk.length
    expect(t).toContain("6"); // {min}
    expect(t).not.toMatch(/min_shot_count/); // audit: the internal code token must never
surface
    expect(t).not.toMatch(/engine/i);
    expect(buildBtn().disabled).toBe(false); // only SOME are at risk ? build can proceed
  });
});

describe("SelectionFooter ? floor warning is jargon-free in kn + hi (#77 regression)", () => {
  beforeEach(() => window.localStorage.clear());
  afterEach(async () => {
    await i18n.changeLanguage("en");
  });

  it("kn: floorOne + floorSome carry no '?????' and no leaked token", async () => {
    await i18n.changeLanguage("kn");
  });
});

```

```

    render(<SelectionFooter {...baseProps} count={1} warning={{ minShotCount: 6, atRisk:
risks(1), allDropped: true }} />);
    expect(warnText()).not.toContain("?????");
    cleanup();
    render(<SelectionFooter {...baseProps} count={5} warning={{ minShotCount: 6, atRisk:
risks(2), allDropped: false }} />);
    expect(warnText()).not.toContain("?????");
    expect(warnText()).not.toMatch(/min_shot_count/);
  });

  it("hi: floorOne + floorSome carry no '????' and no leaked token", async () => {
    await i18n.changeLanguage("hi");
    render(<SelectionFooter {...baseProps} count={1} warning={{ minShotCount: 6, atRisk:
risks(1), allDropped: true }} />);
    expect(warnText()).not.toContain("????");
    cleanup();
    render(<SelectionFooter {...baseProps} count={5} warning={{ minShotCount: 6, atRisk:
risks(2), allDropped: false }} />);
    expect(warnText()).not.toContain("????");
    expect(warnText()).not.toMatch(/min_shot_count/);
  });
});

describe("SelectionFooter ? no-warning controls (summary, reel name, build button)", () => {
  beforeEach(async () => {
    window.localStorage.clear();
    await i18n.changeLanguage("en");
  });

  it("no warning ? no floor-warning region; summary + enabled Build render", () => {
    render(<SelectionFooter {...baseProps} count={4} warning={null} />);
    expect(screen.queryByTestId("floor-warning")).toBeNull();
    expect(screen.getByTestId("selection-summary").textContent).toMatch(/4 rallies/);
    expect(buildBtn().disabled).toBe(false);
  });

  it("building ? the button shows the 'Building?' label and is disabled", () => {
    render(<SelectionFooter {...baseProps} count={2} warning={null} building />);
    expect(buildBtn().textContent).toMatch(/Building/);
    expect(buildBtn().disabled).toBe(true);
  });

  it("empty selection ? Build disabled", () => {
    render(<SelectionFooter {...baseProps} count={0} warning={null} />);
    expect(buildBtn().disabled).toBe(true);
  });

  it("editing the reel name and clicking Build fire their callbacks", () => {
    const onReelNameChange = vi.fn();
    const onBuild = vi.fn();
    render(<SelectionFooter {...baseProps} count={3} warning={null}
onReelNameChange={onReelNameChange} onBuild={onBuild} />);
    fireEvent.change(screen.getByLabelText("Reel name"), { target: { value: "finals" } });
    expect(onReelNameChange).toHaveBeenCalledOnceWith("finals");
    fireEvent.click(buildBtn());
    expect(onBuild).toHaveBeenCalledOnce();
  });
});

```

146. user (2026-06-21T19:26:26.667Z)

File created successfully at: C:/Users/avidu/Projects/khelsutra-cov/web/src/components/SelectionFooter.test.tsx (file state is current in your context ? no need to Read it back)

147. assistant (2026-06-21T19:26:50.573Z)

```
// Content regression guards for the localization catalogs ? the rules the adversarial
translation
// audit (#77) established, pinned so the specific bugs can never silently return. This
complements
// hygiene.test.ts (which proves key/placeholder PARITY) by asserting catalog CONTENT:
// 1. no internal code-token / <code> leaks in any locale,
// 2. the brand "KheIaSutra" is never transliterated into an Indic script,
// 3. engine/server/bucket jargon appears ONLY on keys where the en source uses it,
// 4. the specific flagged mistranslations stay fixed.
// These are pure data assertions over repo-root /locales (D-CATALOG-HOME) ? no render context
needed.
import { describe, it, expect } from "vitest";
import { readFileSync, readdirSync } from "node:fs";
import { join } from "node:path";

const LOCALES_DIR = join(import.meta.dirname, "..", "..", "..", "locales");

type Flat = Record<string, string>;
function flatEntries(obj: Record<string, unknown>, prefix = ""): Flat {
  const out: Flat = {};
  for (const [k, v] of Object.entries(obj)) {
    const key = prefix ? `${prefix}.${k}` : k;
    if (v && typeof v === "object" && !Array.isArray(v)) Object.assign(out, flatEntries(v as
Record<string, unknown>, key));
    else out[key] = String(v);
  }
  return out;
}
const load = (loc: string): Flat =>
  flatEntries(JSON.parse(readFileSync(join(LOCALES_DIR, loc, "common.json"), "utf8")) as
Record<string, unknown>);

const ALL_LOCALES = readdirSync(LOCALES_DIR, { withFileTypes: true })
  .filter((d) => d.isDirectory())
  .map((d) => d.name);
const en = load("en");

describe("catalog content ? no internal token / HTML-tag leaks (every locale)", () => {
  for (const loc of ALL_LOCALES) {
    it(`${loc}: no 'min_shot_count' code token or <code> tag in any user-facing string`, () => {
      const offenders = Object.entries(load(loc))
        .filter(([, v]) => /min_shot_count/.test(v) || /\<\/?code>/i.test(v))
        .map(([k]) => k);
      expect(offenders).toEqual([]);
    });
  }
});

describe("catalog content ? the brand is never transliterated (Indic locales)", () => {
  const TRANSLIT: Record<string, string> = { kn: "?????????", hi: "?????????" };
  for (const loc of ["kn", "hi"] as const) {
    it(`${loc}: no Indic transliteration of "KheIaSutra" anywhere`, () => {
      const offenders = Object.entries(load(loc))
        .filter(([, v]) => v.includes(TRANSLIT[loc]))
        .map(([k]) => k);
      expect(offenders).toEqual([]);
    });
    it(`${loc}: the ingest console keys keep the Latin brand`, () => {
      const e = load(loc);
      for (const key of ["ingest.intro", "ingest.reassure", "ingest.fileHint"]) {
        expect(e[key], `${loc} ${key}`).toContain("KheIaSutra");
      }
    });
  }
});
```

```

    }
  });

describe("catalog content ? engine/server/bucket jargon only where en uses it (Indic locales)",
() => {
  // en deliberately avoids these terms in user-facing body copy (it says "KheIaSutra" / "cloud-
  storage"
  // / passive voice) ? EXCEPT the technical uriAriaLabel ("engine host", "bucket URL"). So the
  rule is
  // self-maintaining: a localized jargon term is allowed ONLY on a key whose en value also uses
  the
  // English word. Reintroducing "engine"/"server"/"bucket" anywhere else fails here.
  const JARGON: Record<string, ReadonlyArray<readonly [string, string]>> = {
    kn: [
      ["?????", "engine"],
      ["?????", "server"],
      ["?????", "bucket"],
    ],
    hi: [
      ["????", "engine"],
      ["????", "server"],
      ["????", "bucket"],
    ],
  };
  for (const loc of ["kn", "hi"] as const) {
    it(`${loc}: a jargon term appears only on keys whose en value also uses it`, () => {
      const e = load(loc);
      const violations: Array<{ key: string; term: string; en: string }> = [];
      for (const [key, val] of Object.entries(e)) {
        for (const [term, eng] of JARGON[loc]) {
          if (val.includes(term) && !(en[key] ?? "").toLowerCase().includes(eng)) {
            violations.push({ key, term, en: en[key] ?? "(absent)" });
          }
        }
      }
      expect(violations).toEqual([]);
    });
  }
});

```

```

describe("catalog content ? flagged mistranslations stay fixed (Kannada/Hindi word choice)", ()
=> {
  it("kn bulk.invert is not the 'rotate' mistranslation (????????)", () => {
    expect(load("kn")["bulk.invert"]).not.toBe("????????");
  });
  it("kn result.dismiss is not the legalistic '????????'", () => {
    expect(load("kn")["result.dismiss"]).not.toBe("????????");
  });
  it("hi result.dismiss is not the harsh '????? ?????'", () => {
    expect(load("hi")["result.dismiss"]).not.toBe("????? ?????");
  });
});

```

148. user (2026-06-21T19:26:50.603Z)

File created successfully at: C:/Users/avidu/Projects/khelsutra-cov/web/src/i18n/catalog-invariants.test.ts (file state is current in your context ? no need to Read it back)

149. assistant (2026-06-21T19:26:53.474Z)

Now run coverage to see the new numbers and confirm the thresholds are satisfied:

150. user (2026-06-21T19:27:05.068Z)

```

> khelsutra-web@0.0.0 test:coverage
> vitest run --coverage
  Coverage enabled with v8
  ? src/i18n/catalog-invariants.test.ts (15 tests)
25ms
  ? src/i18n/hygiene.test.ts (28 tests)
32ms
  ? src/components/QueryBar.examples.test.ts (2 tests)
16ms
  ? src/lib/errorText.test.ts (5 tests)
52ms
  ? src/components/SelectionFooter.test.tsx (9 tests)
732ms
  ? SelectionFooter ? floor warning renders the audit-clean copy (en) >
allDropped + a single selection ? floorOne, passive phrasing (no 'engine'), Build blocked
454ms
  ? src/i18n/i18n.test.tsx (17 tests)
1637ms
  ? i18n scaffold (B1) > renders user-facing copy from the en
catalog (not raw keys) 717ms
  Test Files 22 passed (22)
  Tests 201 passed (201)
  % Coverage report from v8
All files | 92.61 | 87.74 | 82.35 | 92.61 |
App.tsx | 74.89 | 78.04 | 25 | 74.89 | ...10-215,242-244
QueryBar.tsx | 86.84 | 76.92 | 40 | 86.84 | ...41-43,68-70,82
src/i18n | 100 | 25 | 100 | 100 |
index.ts | 100 | 25 | 100 | 100 | 72-76
errorText.ts | 100 | 100 | 100 | 100 |
===== Coverage summary =====
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

151. assistant (2026-06-21T19:28:07.818Z)

Coverage improved: ****92.61% stmts / 87.74% branch / 82.35% funcs / 92.61% lines**** (was 90.44/86.86/80.45/90.44), 201 tests pass (+24), and `SelectionFooter` is now fully covered. The thresholds pass with margin. Let me tighten the floor to just below current (a real non-regression ratchet, but not brittle):

152. user (2026-06-21T19:28:11.085Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/vite.config.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

153. assistant (2026-06-21T19:28:24.905Z)

I added one extra brace in that edit. Let me read the end of the file and fix the structure:

154. user (2026-06-21T19:28:25.661Z)

```

24      // thresholds, so untested new code is caught. `all` counts every source file (not
only the ones a
25      // test imported), so deleting a test surfaces here too. Excludes are non-logic
files: the app
26      // entry/bootstrap, type-only declarations, and the test harness itself.
27      coverage: {
28        provider: "v8",
29        reporter: ["text", "text-summary"],
30        all: true,
31        include: ["src/**/*.{ts,tsx}"],
32        exclude: [
33          "src/**/*.test.{ts,tsx}", // the tests
34          "src/test/**", // jsdom localStorage shim (test harness, not shipped logic)
35          "src/main.tsx", // app entry: mounts <App/> + bundles fonts ? nothing to unit-

```

```

test
36         "src/api/types.ts", // type-only declarations (no runtime code)
37         "**/*.d.ts",
38     ],
39     // Floor set just below current (stmts/lines 92.6, branches 87.7, funcs 82.4) ? a
real ratchet
40     // that fails CI on a meaningful drop, with a little slack so it doesn't flap.
Ratchet upward as
41     // coverage grows. (functions is kept looser: App.tsx is the engine-orchestrating
shell, costly
42     // to exercise fully in unit tests.)
43     thresholds: {
44         statements: 92,
45         branches: 87,

```

155. user (2026-06-21T19:28:29.945Z)

```

43         thresholds: {
44             statements: 92,
45             branches: 87,
46             functions: 80,
47             lines: 92,
48         },
49     },
50 },
51 },
52 });
53

```

156. assistant (2026-06-21T19:28:33.674Z)

Confirmed ? line 50 is the extra brace. Removing it:

157. user (2026-06-21T19:28:35.053Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/vite.config.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)

158. assistant (2026-06-21T19:28:38.226Z)

Now wire CI to enforce the floor (web job runs `test:coverage` instead of `test`):

159. user (2026-06-21T19:28:39.681Z)

```

31     web:
32         # Operator-console SPA (web/): typecheck + unit tests + production build gate the
merge.
33         runs-on: ubuntu-latest
34         defaults:
35             run:
36                 working-directory: web
37         steps:
38             - uses: actions/checkout@v4
39             - uses: actions/setup-node@v4
40               with:
41                 node-version: "24"
42                 cache: "npm"
43                 cache-dependency-path: web/package-lock.json
44             - run: npm ci
45             - run: npm run typecheck
46             - run: npm run test
47             - run: npm run build

```

160. user (2026-06-21T19:28:47.473Z)

The file C:/Users/avidu/Projects/khelsutra-cov/.github/workflows/test.yml has been updated successfully. (file state is current in your context ? no need to Read it back)

161. assistant (2026-06-21T19:28:54.819Z)

Now full local verification ? typecheck, `test:coverage` (must pass the floor), and the production build:

162. user (2026-06-21T19:29:12.612Z)

```
=== TYPECHECK ===
> khelsutra-web@0.0.0 typecheck
> tsc --noEmit

src/components/SelectionFooter.test.tsx(13,1): error TS6133: 'FloorRisk' is declared but its
value is never read.
=== TEST:COVERAGE ===
    Coverage enabled with v8
    Test Files   22 passed (22)
    Tests       201 passed (201)
File           | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
All files      | 92.61   | 87.74    | 82.35   | 92.61   |
=== BUILD ===
dist/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
  27.58 kB
dist/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
  90.80 kB
dist/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
 121.19 kB
dist/assets/index-Dt-eK0X5.css
  24.85 kB ? gzip: 5.43 kB
dist/assets/index-CICa-Suj.js
 409.85 kB ? gzip: 124.11 kB
? built in 1.27s
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

163. assistant (2026-06-21T19:29:21.586Z)

Typecheck caught an unused `FloorRisk` import (the warning literals are already type-checked via the prop). Removing it:

164. user (2026-06-21T19:29:23.478Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/src/components/SelectionFooter.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

165. user (2026-06-21T19:29:31.175Z)

```
=== TYPECHECK ===

> khelsutra-web@0.0.0 typecheck
> tsc --noEmit

exit: 0
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

166. assistant (2026-06-21T19:30:18.295Z)

Typecheck clean; coverage passes the floor; build succeeds. Per the ultracode mandate, before merging I'll ****adversarially verify**** the regression tests are non-tautological ? i.e., empirically prove each one ***fails*** if the audit bug is reintroduced. Let me run mutation tests (mutate catalog ? run test ? expect failure ? revert):

167. user (2026-06-21T19:30:41.844Z)

MUTATION 1: reintroduce '?????' (engine) into kn footer.floorOne

```
catalog-invariants:
  Tests 1 failed | 14 passed
(15)
SelectionFooter:
  Tests 1 failed | 8 passed
(9)
reverted kn
```

MUTATION 2: reintroduce '<code>min_shot_count</code>' into hi footer.floorSomeRest

```
catalog-invariants:
  Tests 1 failed | 14 passed
(15)
reverted hi
```

MUTATION 3: revert hi brand to ???????? in ingest.intro

```
catalog-invariants:
  Tests 2 failed | 13 passed
(15)
reverted hi
```

POST-REVERT cleanliness (should be empty):

Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

168. assistant (2026-06-21T19:31:09.353Z)

The mutation tests prove the guards are **non-tautological**: every reintroduced audit bug fails a test (engine? both data + render tests fail; ``` token ? fails; brand transliteration ? 2 tests fail), and ``/locales`` reverted clean.

Let me verify the changeset (especially that ``package-lock.json`` captured the coverage dep so CI's ``npm ci`` works), then run an adversarial review workflow over the PR.

169. user (2026-06-21T19:31:10.858Z)

```
=== changed/new files vs origin/master ===
M .github/workflows/test.yml
M web/package-lock.json
M web/package.json
M web/vite.config.ts
?? web/src/components/SelectionFooter.test.tsx
?? web/src/i18n/catalog-invariants.test.ts
=== package-lock includes coverage-v8? ===
3
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

170. assistant (2026-06-21T19:31:22.871Z)

``package-lock.json`` captured the coverage dep (CI ``npm ci`` will install it); changeset is 6 files, scoped to ``web/`` + CI. Now the adversarial review ? fan out independent reviewers across distinct lenses, then synthesize:

171. user (2026-06-21T19:32:10.650Z)

Workflow launched in background. Task ID: wsto4ws8s
Summary: Adversarial multi-lens review of the web coverage-floor + i18n regression-test PR
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\54d4074a-5a8d-4058-b42d-a757c338132c\subagents\workflows\wf_1c9ea2fd-eb8
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\54d4074a-5a8d-4058-b42d-a757c338132c\workflows\scripts\review-coverage-pr-wf_1c9ea2fd-eb8.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\54d4074a-5a8d-4058-b42d-a757c338132c\workflows\scripts\review-coverage-pr-wf_1c9ea2fd-eb8.js"} to iterate without resending the script.)
Run ID: wf_1c9ea2fd-eb8
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--

Users-avidu-Projects-khelsutra\54d4074a-5a8d-4058-b42d-a757c338132c\workflows\scripts\review-coverage-pr-wf_1c9ea2fd-eb8.js", resumeFromRunId: "wf_1c9ea2fd-eb8"}}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

172. assistant (2026-06-21T19:32:26.580Z)

The adversarial review is running in the background (5 parallel reviewers + synthesis). Here's where things stand while it completes:

```
**Done so far (in worktree `khelsutra-cov`, branch `chore/web-coverage-floor`):**
- Stood up the repo's **first coverage measurement** for `web` (`@vitest/coverage-v8` + `test:coverage` script + thresholds in `vite.config.ts`) and wired CI to enforce it (`test.yml` web job ? `npm run test:coverage`).
- Added 2 test files (+24 tests): `SelectionFooter.test.tsx` (renders the previously-uncovered floor-warning block in en/kn/hi) and `catalog-invariants.test.ts` (locks the #77 audit rules repo-wide).
- **Coverage improved: 90.44% ? 92.61% lines** (stmts 92.61, branch 87.74, funcs 82.35); 201 tests green; typecheck + build clean.
- **Mutation-tested** the guards: every reintroduced audit bug (engine jargon, `` token, brand transliteration) fails a test ? proven non-tautological.
```

I'll apply any must-fix findings from the review, then commit ? PR ? merge on green CI. Awaiting the workflow result.

173. user (2026-06-21T19:37:44.800Z)

```
<task-notification>
<task-id>wsto4ws8s</task-id>
<tool-use-id>toolu_01367Nm5v1nxhR5BjGVZ2eoZ</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-
khelsutra\54d4074a-5a8d-4058-b42d-a757c338132c\tasks\wsto4ws8s.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Adversarial multi-lens review of the web coverage-floor + i18n
regression-test PR" completed</summary>
<result>{"synthesis":{"overallVerdict":"ship-with-
fixes","mustFixBeforeMerge":[{"title":"catalog-invariants has NO guard for #77 class (d) Hindi
over-Sanskritization ? reverting the buggy terms keeps the suite
green","file":"C:/Users/avidu/Projects/khelsutra-cov/web/src/i18n/catalog-
invariants.test.ts","fix":"I empirically reverted locales/hi/common.json grid.shots -&gt;
'~{count} ????? · ??????????' and grid.shotsConfidence -&gt; ' ? ??????: {conf}' and ran the
new tests: all 24 (15 catalog-invariants + 9 SelectionFooter) PASS. The exact #77 bug returns
undetected because none of ??????????/?????????/??????? appears anywhere in either test file
(grep count = 0), and these keys never render through SelectionFooter. Add a positive assertion
that locks the cleaned replacements on the audited keys, e.g.
expect(load('hi')['grid.shots']).toContain('????????');
expect(load('hi')['grid.shotsConfidence']).toContain('????? ?????');
expect(load('hi')['query.support']).toContain('????????'); and a banned-term scan: const
HI_BANNED=['?????????', '?????????', '?????????']; for each term assert no hi entry contains it.
Without this the file cannot honestly be credited as a #77 lock ? it ships false reassurance for
class (d)."}, {"title":"Brand guard positively checks only 3 of the 10 en keys that carry
'KhelSutra' ? ingest.hint, errors.networkError, errors.invalidUri are unguarded (all three were
fixed by #77)","file":"C:/Users/avidu/Projects/khelsutra-cov/web/src/i18n/catalog-
invariants.test.ts","fix":"The positive check loops a hardcoded
['ingest.intro', 'ingest.reassure', 'ingest.fileHint'], but en contains 'KhelSutra' on 10 keys
(verified): the 3 checked plus ingest.hint, errors.networkError, errors.invalidUri (SPA keys #77
fixed) and 4 site.* keys. Make it self-maintaining: for every en entry whose value includes
'KhelSutra', assert the kn AND hi value for that same key also includes 'KhelSutra'. This closes
ingest.hint + both errors.* in one rule and auto-covers future brand keys."}, {"title":"Brand no-
transliteration guard is a single-spelling denylist ? alternate transliterations evade
it","file":"C:/Users/avidu/Projects/khelsutra-cov/web/src/i18n/catalog-
invariants.test.ts","fix":"The negative guard denylists exactly ???????? (hi) / ?????????? (kn).
I injected the spaced form '?? ???? ? ? ? ? ?' into the unguarded ingest.hint and all 15
invariant tests stayed green ? the brand was transliterated and the test passed. The positive-
```

presence rule from the previous fix (require Latin 'KheLSutra' on every key where en has it) structurally subsumes this and catches all spellings at once; adopt it instead of relying on the one-spelling denylist."}], "optionalFollowups": ["Convert the SelectionFooter kn/hi/en jargon negatives (not.toMatch(/?????/ /????/ /min_shot_count/ /engine/i)) into POSITIVE render assertions ? they are tautological today (those tokens live only on ingest.uriAriaLabel, which the footer never renders, so they are always-true). Assert the localized floor copy actually renders (e.g. '2 ?????????' / '2 ?????????'), which exercises the kn/hi catalog + ICU plural path instead of pinning a state the code cannot produce.", "Replace the floorSomeLead toContain('2') pin with toMatch(/2 rallies/) (en) so the ICU plural 'other' arm is verified, not just digit presence.", "Switch the flagged-mistranslation pins from full-string .not.toBe to substring .not.toContain (e.g. not.toContain('????????'), not.toContain('????????'), not.toContain('?????')) so a padded/wrapped variant of the banned term ('????? ?????????', trailing space) is still caught.", "Broaden the jargon scan from 3 fixed transliterations to also denylist Latin tech tokens (engine|server|bucket|backend) inside non-en values on keys where en omits them, and add known synonyms (????? / ?????????, ?????? / ??????); generalize the <code> tag check to /<\\/?(code|kbd|samp|tt|var)\\b/i. At minimum document that the guard is term-bound, not concept-bound.", "Add web/src/components/QueryBar.test.tsx ? the cheapest remaining real coverage win (QueryBar.test.tsx is funcs 40% / lines 86.84%, untested handlers applyText/submit + pill branches). A ~30-line render test mirroring SelectionFooter.test.tsx would lift the loosest metric (functions, 82.35 vs 80 floor).", "Tighten the functions threshold from 80 toward 82 (the loosest gate, +2.35 slack) once the QueryBar test lands, so the metric App.tsx protects least gets a real floor; keep the App.tsx-cost rationale in the comment.", "In the PR description, state the coverage split honestly: the excludes (main.tsx + api/types.ts) account for ~91.36 baseline; only SelectionFooter.test.tsx adds the real ~1.25 to 92.61, and catalog-invariants.test.ts adds ZERO source-line coverage (it is a content-regression guard, not part of the ratchet). Don't frame the full 90.44->92.61 as new test coverage.", "Optionally assert key presence (expect(e).toHaveProperty(key)) before toContain in the brand/ingest checks so a renamed key yields a readable failure rather than a TypeError on undefined.", "Commit the work: vite.config.ts shows as modified and both test files are untracked on chore/web-coverage-floor. Stage explicit paths (git add .github/workflows/test.yml web/package.json web/package-lock.json web/vite.config.ts web/src/components/SelectionFooter.test.tsx web/src/i18n/catalog-invariants.test.ts) per the concurrent-checkout convention."], "summary": "The coverage/CI machinery is sound and shippable; the #77 regression-net guard is incomplete and must be tightened before it can be honestly credited as a #77 lock ? hence ship-with-fixes rather than ship.\\n\\n\\nVERIFIED GREEN (the four 'ship'/'ship-with-fixes' votes on config + CI are correct): I ran the full coverage suite locally ? statements 92.61 / branches 87.74 / functions 82.35 / lines 92.61, all at or above the configured floors (92/87/80/92), so CI passes on merge. The excludes (main.tsx, api/types.ts, test/**, *.d.ts) are correctly scoped to non-logic files; all:true + include src/**/*.{ts,tsx} makes new untested files drag the ratio down. test.yml runs npm run test:coverage, @vitest/coverage-v8 ^3.2.6 is in devDeps + lockfile (npm ci clean), and Node 24 satisfies the vitest 3.2.6 engines. The site job is untouched. SelectionFooter.test.tsx is a genuinely strong addition (drives SelectionFooter.test.tsx 66->100%).\\n\\n\\nWHY NOT 'ship' ? the lone 'needs-work' blocker is REAL and I reproduced it: the new catalog-invariants.test.ts claims to lock all of #77, but #77 class (d) (Hindi over-Sanskritization) has zero guard. Reverting locales/hi/common.json grid.shots back to '?????????' and grid.shotsConfidence to '???????' leaves all 24 new tests green ? the exact original bug sails back in silently. I also confirmed two major denylist gaps: the brand positive-check covers only 3 of the 10 en keys carrying 'KheLSutra' (ingest.hint + errors.networkError + errors.invalidUri, all #77 fixes, are unguarded), and the no-transliteration check is a one-spelling denylist that a spaced '??? ?????' injected into ingest.hint evades (15/15 still green). The single fix ? derive guards positively from en's own values (presence of the cleaned hi replacements; presence of Latin 'KheLSutra' wherever en has it) ? closes all three at once and makes the file self-maintaining.\\n\\n\\nLower-priority but worth noting: several SelectionFooter render-negatives (not.toMatch /engine/, /min_shot_count/, /?????/) are tautological because those tokens live only on a key the footer never renders, so they are always-true confidence theater rather than real render coverage. None of this affects the coverage number; the gate itself is safe to merge once the guard gaps are closed."}, "reviews": [{"dimension": "coverage-config", "verdict": "ship", "findings": [{"severity": "praise", "title": "Excludes are correctly scoped to genuinely non-testable files ? verified by opening each", "detail": "I opened every excluded path and confirmed none hides runtime logic the gate should"}]}], "truncated": 33953, "fullResult": "C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-kheLSutra\\54d4074a-5a8d-4058-b42d-a757c338132c\\tasks\\wsto4ws8s.output", "usage": {"agent_count": 6, "subagent_tokens": 320262, "tool_uses": 84, "duration_ms": 334061}}

</task-notification>

174. assistant (2026-06-21T19:38:40.845Z)

The review came back **ship-with-fixes** with a genuinely valuable, *empirically reproduced* blocker and two major gaps in my `catalog-invariants.test.ts`. Per my own "verify, don't trust claims" discipline, let me reproduce the key gap myself, enumerate the brand keys, and read `QueryBar.tsx` (for a suggested coverage win) before fixing.

175. user (2026-06-21T19:38:49.434Z)

```
=== REPRODUCE class-(d) gap: revert hi grid.shotsConfidence to ???????, run catalog-invariants
===
    Tests 15 passed (15)
reverted hi
=== en keys containing the Latin brand 'KheI Sutra' ===
ingest.intro
ingest.hint
ingest.reassure
ingest.fileHint
errors.networkError
errors.invalidUri
site.hero.lead
site.record.intro
site.record.upload.body
site.capture.step4.b2
total: 10
Shell cwd was reset to C:\Users\avidu\Projects\kheIsutra
```

176. assistant (2026-06-21T19:39:05.909Z)

Both gaps confirmed real: class-(d) revert stays green (15 passed), and en carries "KheI Sutra" on **10** keys (I only guarded 3). The self-maintaining rule must hold on current catalogs ? let me verify it (and the class-(d) terms) before encoding, so the new tests are green:

177. user (2026-06-21T19:39:17.020Z)

```
=== self-maintaining brand rule: every en-brand key has Latin brand in kn AND hi ===
    ALL 10 keys OK in kn+hi

=== class-(d) banned Hindi terms absent? ===
    ??????????: absent ?
    ??????????: absent ?
    ??????: absent ?

=== class-(d) cleaned replacements present? ===
    hi grid.shots has ??????: true
    hi grid.shotsTitle has ??????: true
    hi query.support has ??????: true
    hi grid.shotsConfidence has ?????: true

=== broadened jargon: Latin tech tokens in non-en where en omits? (should be none on rendered keys) ===
    (end Latin-jargon scan)
Shell cwd was reset to C:\Users\avidu\Projects\kheIsutra
```

178. assistant (2026-06-21T19:40:19.788Z)

Every upgraded invariant holds on current catalogs (brand rule passes all 10 keys; class-(d) terms absent; replacements present; no Latin-jargon leaks). The blocker is real and reproduced. Now let me read `QueryBar.tsx` (the suggested coverage win) so I can address that follow-up too, then rewrite the tests comprehensively.

179. user (2026-06-21T19:40:20.334Z)

```

1      // The fast-entry query bar (PER_RALLY_SELECTION.md P3). A deterministic, OFFLINE parser
turns
2      // plain text ("top 15 longest", "over 10s") into a selection over the SAME shared set
the
3      // checkboxes use. It is honest by construction: a live "parsed-pill" shows exactly how
the
4      // text was understood, and concepts with no data behind them (shot-type / player /
score) are
5      // greyed out as "not available yet" ? never silently faked.
6      import { useMemo, useState } from "react";
7      import type { FormEvent } from "react";
8      import { useTranslation, Trans } from "react-i18next";
9      import { parseQuery, summarize } from "../lib/query";
10     import type { ParsedQuery } from "../lib/query";
11
12     interface Props {
13       /** Apply a parsed query to the rally set (App runs runQuery + writes the shared
selection).
14       * `source` distinguishes a typed-and-submitted query from an example-chip tap (for
the trace). */
15       onApply: (q: ParsedQuery, source: "typed" | "example") => void;
16     }
17
18     // Each example is a real, parseable query ? tapping one fills the box AND applies it.
Led by rally
19     // LENGTH (the trusted axis); shot-count filtering exists but is experimental, so it's
not featured
20     // here (a test pins that every example parses to a recognized, fully-available query).
21     export const EXAMPLES = ["over 10s", "top 15 longest", "shortest first", "first 10"];
22
23     // Parsed-pill chips: legible on a phone (text-sm, not text-xs) ? A4.
24     const CHIP = "rounded-full px-3 py-1 text-sm font-medium";
25
26     export function QueryBar({ onApply }: Props) {
27       const { t, i18n } = useTranslation();
28       // The example chips are ENGLISH query syntax (the parser is English-only). For a non-
English user
29       // they're an untappable wall ? hide them and let the localized TapFilters (A9) own
the tap path.
30       const isEnglish = (i18n.resolvedLanguage ?? i18n.language ?? "en").split("-")[0] ===
"en";
31       const [value, setValue] = useState("");
32       const parsed = useMemo(() => parseQuery(value), [value]);
33       const clauses = summarize(parsed);
34       const typed = value.trim().length > 0;
35
36       const applyText = (text: string) => {
37         setValue(text);
38         onApply(parseQuery(text), "example");
39       };
40       const submit = (e: FormEvent) => {
41         e.preventDefault();
42         if (parsed.recognized) onApply(parsed, "typed");
43       };
44
45       return (
46         <div>
47           <form onSubmit={submit} className="flex flex-wrap items-center gap-2">
48             <input
49               type="text"
50               value={value}
51               onChange={(e) => setValue(e.target.value)}
52               placeholder={t("query.placeholder")}
53               aria-label={t("query.ariaLabel")}
54               className="min-h-11 min-w-[16rem] flex-1 rounded-xl border border-black/15 bg-

```

```

white px-4 py-2 text-sm text-ink outline-none focus:border-green"
55         />
56         <button
57             type="submit"
58             disabled={!parsed.recognized}
59             className="min-h-11 rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-
white disabled:opacity-40"
60         >
61             {t("query.apply")}
62         </button>
63     </form>
64
65     {/* Live interpretation ? transparent + hand-correctable (no hidden NL guessing).
*/}
66     <div aria-live="polite" data-testid="query-pill" className="mt-2 flex flex-wrap
items-center gap-1.5">
67         {clauses.map((c) => (
68             <span key={c} className={CHIP + " bg-green/15 text-green"}>
69                 {c}
70             </span>
71         ))}
72         {parsed.unavailable.map((u) => (
73             <span
74                 key={u.token}
75                 title={t("query.notAvailableTitle", { token: u.token, kind: u.kind })}
76                 className={CHIP + " bg-black/5 text-ink/45"}
77             >
78                 <span className="line-through">{u.token}</span> · {t("query.notAvailable")}
79             </span>
80         ))}
81         {typed && !parsed.recognized && parsed.unavailable.length === 0 && (
82             <span className="text-xs text-ink/50">{t("query.unreadable")}</span>
83         )}
84         {!typed && <span className="text-xs text-ink/50">{t("query.hint")}</span>}
85     </div>
86
87     {/* Surface the "not available yet" reason as VISIBLE text (the per-chip title is
hover-only,
88         so it never appears on a phone) ? A4. */}
89     {parsed.unavailable.length > 0 && (
90         <p className="mt-1.5 text-xs text-ink/50">{t("query.notAvailableHelp")}</p>
91     )}
92
93     {isEnglish && (
94         <div className="mt-3 flex flex-wrap items-center gap-2 text-sm">
95             <span className="text-ink/40">{t("query.try")}</span>
96             {EXAMPLES.map((ex) => (
97                 <button
98                     key={ex}
99                     type="button"
100                     onClick={() => applyText(ex)}
101                     className="inline-flex min-h-11 items-center rounded-full border border-
black/15 px-4 text-sm hover:border-green"
102                 >
103                     {ex}
104                 </button>
105             ))}
106         </div>
107     )}
108
109     <p className="mt-2 text-xs text-ink/40">
110         <Trans i18nKey="query.support" components={{ b: <strong className="text-ink/60"
/>, x: <span className="text-amber-700" /> }} />
111     </p>
112 </div>

```

```
113     );
114   }
115
```

180. assistant (2026-06-21T19:40:54.395Z)

QueryBar's uncovered paths are the `submit`/`applyText` handlers and the pill/unreadable branches ? a clean coverage win. I need a known unrecognized-with-no-unavailable input for the "unreadable" branch; let me check the query lib's existing tests for a safe one.

181. user (2026-06-21T19:41:02.654Z)

```
41:   expect(parseQuery("over 10s").filters).toEqual([{ field: "duration", op: ">", value: 10
  }]);
44:   expect(parseQuery("longer than 10 seconds").filters[0]).toEqual({ field: "duration", op:
  ">", value: 10 });
45:   expect(parseQuery("under 5 sec").filters[0]).toEqual({ field: "duration", op: "<", value:
  5 });
46:   expect(parseQuery("at least 8s").filters[0]).toEqual({ field: "duration", op: ">=",
  value: 8 });
47:   expect(parseQuery("at most 8 seconds").filters[0]).toEqual({ field: "duration", op: "<=",
  value: 8 });
48:   expect(parseQuery("exactly 7s").filters[0]).toEqual({ field: "duration", op: "=", value:
  7 });
51:   expect(parseQuery("over 1 minute").filters[0]).toEqual({ field: "duration", op: ">",
  value: 60 });
52:   expect(parseQuery("at least 1.5m").filters[0]).toEqual({ field: "duration", op: ">=",
  value: 90 });
55:   expect(parseQuery("? 10s").filters[0]).toEqual({ field: "duration", op: ">=", value: 10
  });
56:   expect(parseQuery("> 10").filters[0]).toEqual({ field: "duration", op: ">", value: 10 });
57:   expect(parseQuery("10s+").filters[0]).toEqual({ field: "duration", op: ">=", value: 10
  });
58:   expect(parseQuery("10s or longer").filters[0]).toEqual({ field: "duration", op: ">=",
  value: 10 });
59:   expect(parseQuery("8s or less").filters[0]).toEqual({ field: "duration", op: "<=", value:
  8 });
62:   expect(parseQuery("no more than 5s").filters[0]).toEqual({ field: "duration", op: "<=",
  value: 5 });
65:   expect(parseQuery("10s").filters).toEqual([]);
66:   expect(parseQuery("10s").recognized).toBe(false);
72:   expect(parseQuery("over 10 shots").filters[0]).toEqual({ field: "shot_count", op: ">",
  value: 10 });
73:   expect(parseQuery("at least 8 shots").filters[0]).toEqual({ field: "shot_count", op:
  ">=", value: 8 });
74:   expect(parseQuery("fewer than 5 shots").filters[0]).toEqual({ field: "shot_count", op:
  "<", value: 5 });
75:   expect(parseQuery("10+ shots").filters[0]).toEqual({ field: "shot_count", op: ">=",
  value: 10 });
=== quick parser probe via a temp test ===
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

182. assistant (2026-06-21T19:41:11.833Z)

The probe output didn't capture. Existing test (line 66) proves `parseQuery("10s").recognized === false` with empty filters ? likely the cleanest "unreadable" input. Let me confirm its `unavailable` is empty:

183. user (2026-06-21T19:41:17.802Z)

```
PROBE over 10s | recognized: true | unavailable: [] | clauses: ["length >
10s"]
PROBE 10s | recognized: false | unavailable: [] | clauses: []
PROBE zzzzz | recognized: false | unavailable: [] | clauses: []
```

(cleaned up)
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

184. assistant (2026-06-21T19:42:18.910Z)

Both inputs confirmed (`over 10s`?recognized, `10s`?unreadable, both `unavailable:[]`). Now I'll apply all the review fixes. First, the comprehensive rewrite of `catalog-invariants.test.ts` (closes the blocker + both majors + the broadenings):

185. assistant (2026-06-21T19:42:48.240Z)

```
// Content regression guards for the localization catalogs ? the rules the adversarial
translation
// audit (#77) established, pinned so the specific bugs can never silently return. This
complements
// hygiene.test.ts (which proves key/placeholder PARITY) by asserting catalog CONTENT. Each
block maps
// to a concrete #77 bug class; the mutation that reintroduces the bug fails here. Pure data
assertions
// over repo-root /locales (D-CATALOG-HOME) ? no render context needed.
import { describe, it, expect } from "vitest";
import { readFileSync, readdirSync } from "node:fs";
import { join } from "node:path";

const LOCALES_DIR = join(import.meta.dirname, "..", "..", "..", "locales");

type Flat = Record<string, string>;
function flatEntries(obj: Record<string, unknown>, prefix = ""): Flat {
  const out: Flat = {};
  for (const [k, v] of Object.entries(obj)) {
    const key = prefix ? `${prefix}.${k}` : k;
    if (v && typeof v === "object" && !Array.isArray(v)) Object.assign(out, flatEntries(v as
Record<string, unknown>, key));
    else out[key] = String(v);
  }
  return out;
}
const load = (loc: string): Flat =>
  flatEntries(JSON.parse(readFileSync(join(LOCALES_DIR, loc, "common.json"), "utf8")) as
Record<string, unknown>);

const ALL_LOCALES = readdirSync(LOCALES_DIR, { withFileTypes: true })
  .filter((d) => d.isDirectory())
  .map((d) => d.name);
const en = load("en");

// (1) Internal-token / debug-tag leaks ? EVERY locale. The engine's internal `min_shot_count`
config
// name leaked into footer.floorSomeRest (#77, kn even wrapped it in <code>). No
developer/markup tag
// (code/kbd/samp/tt/var) or that token may reach any user-facing string.
describe("catalog content ? no internal token / debug-tag leaks (every locale)", () => {
  const TOKEN = /min_shot_count/;
  const DEBUG_TAG = /<\?/(?:(code|kbd|samp|tt|var)\b/i;
  for (const loc of ALL_LOCALES) {
    it(`${loc}: no 'min_shot_count' token or <code>/<kbd>/<samp>/<tt>/<var> tag`, () => {
      const offenders = Object.entries(load(loc))
        .filter(([k, v]) => TOKEN.test(v) || DEBUG_TAG.test(v))
        .map([k] => k);
      expect(offenders).toEqual([]);
    });
  }
});

// (2) Brand ? never transliterated, and present (Latin) on EVERY key en uses it on. The
```



```

positive rule
// is the real guard: self-maintaining (auto-covers future brand keys, incl. errors.* /
ingest.hint that
// the old 3-key denylist missed) and it catches ANY transliteration spelling at once ? a
transliterated
// value simply won't contain Latin "KheIaSutra". The Indic-spelling denylist is cheap defence-
in-depth.
describe("catalog content ? the brand stays Latin 'KheIaSutra' everywhere en uses it", () => {
  const brandKeys = Object.keys(en).filter((k) => en[k].includes("KheIaSutra"));

  it("en actually carries the brand on several keys (so the guard is meaningful)", () => {
    expect(brandKeys.length).toBeGreaterThanOrEqual(6);
  });

  for (const loc of ["kn", "hi"] as const) {
    it(`${loc}: Latin "KheIaSutra" is present on every key whose en value uses it`, () => {
      const e = load(loc);
      const missing = brandKeys.filter((k) => !(e[k] ?? "").includes("KheIaSutra"));
      expect(missing).toEqual([]);
    });
  }
}

const TRANSLIT: Record<string, string> = { kn: "?????????", hi: "?????????" };
for (const loc of ["kn", "hi"] as const) {
  it(`${loc}: the known Indic transliteration of the brand never appears`, () => {
    const offenders = Object.entries(load(loc))
      .filter(([v]) => v.includes(TRANSLIT[loc]))
      .map(([k]) => k);
    expect(offenders).toEqual([]);
  });
}
});

// (3) Engine/server/bucket/backend jargon only where en uses it (kn/hi). Self-maintaining: a
localized
// jargon term ? the Latin word OR an Indic transliteration ? is allowed ONLY on a key whose en
value
// also uses the English word. en avoids these in body copy (uses the brand / "cloud-storage" /
passive
// voice) EXCEPT the technical ingest.uriAriaLabel + uriPlaceholder; #77 leaked them into kn/hi
elsewhere.
describe("catalog content ? engine/server/bucket jargon only where en uses it (kn/hi)", () => {
  const ENGLISH = ["engine", "server", "bucket", "backend"] as const;
  const TRANSLIT: Record<string, Record<(typeof ENGLISH)[number], string[]>> = {
    kn: { engine: ["?????"], server: ["?????", "?????"], bucket: ["?????"], backend:
["?????????"] },
    hi: { engine: ["????"], server: ["?????", "????????"], bucket: ["????"], backend: ["????????"]
},
  };
  for (const loc of ["kn", "hi"] as const) {
    it(`${loc}: a jargon term (Latin or transliterated) appears only where en uses the English
word`, () => {
      const e = load(loc);
      const violations: Array<{ key: string; term: string; en: string }> = [];
      for (const [key, val] of Object.entries(e)) {
        const enUsesWord = (w: string) => new RegExp(`\\b${w}\\b`, "i").test(en[key] ?? "");
        for (const w of ENGLISH) {
          const present = [
            ...(new RegExp(`\\b${w}\\b`, "i").test(val) ? [w] : []),
            ...TRANSLIT[loc][w].filter((t) => val.includes(t)),
          ];
          for (const term of present) if (!enUsesWord(w)) violations.push({ key, term, en:
en[key] ?? "(absent)" });
        }
      }
    }
  }
}

```

```

    expect(violations).toEqual([]);
  });
}
});

// (4) Hindi de-Sanskritization (#77 class d) ? the over-Sanskritized / stiff terms must stay
replaced
// on the AUDITED keys. Key-scoped (not a repo-wide ban) so legitimate marketing prose elsewhere
is not
// constrained, while still catching a silent revert of the exact bug.
describe("catalog content ? Hindi de-Sanskritization stays fixed (#77 class d)", () => {
  const hi = load("hi");
  const BANNED: Array<[string, string]> = [
    ["grid.shots", "?????????", // "experimental" calque
    ["grid.shotsTitle", "?????????"],
    ["grid.shotsTitle", "?????????", // "under development" bureaucratese
    ["query.support", "?????????"],
    ["query.support", "?????????"],
    ["grid.shotsConfidence", "?????????", // "faith" ? wrong sense for statistical confidence
    ["app.badge", "?????-????", // "per-rally" calque
  ];
  for (const [key, term] of BANNED) {
    it(`hi ${key} no longer contains the stiff "${term}"`, () => {
      expect(hi).toHaveProperty(key);
      expect(hi[key] ?? "").not.toContain(term);
    });
  }
  it("hi keeps the friendlier replacements (??????? / ????? ?????)", () => {
    expect(hi["grid.shots"]).toContain("????????");
    expect(hi["grid.shotsTitle"]).toContain("????????");
    expect(hi["query.support"]).toContain("????????");
    expect(hi["grid.shotsConfidence"]).toContain("????? ?????");
  });
});

// (5) Flagged mistranslations ? substring guard (.not.toContain) so a padded/wrapped variant of
the
// banned term ("????? ?????????", a trailing space, ?) is still caught.
describe("catalog content ? flagged mistranslations stay fixed (kn/hi word choice)", () => {
  it("kn bulk.invert drops the 'rotate' mistranslation (????????)", () => {
    expect(load("kn")["bulk.invert"]).not.toContain("????????");
  });
  it("kn result.dismiss drops the legalistic '????????'", () => {
    expect(load("kn")["result.dismiss"]).not.toContain("????????");
  });
  it("hi result.dismiss drops the harsh '?????'", () => {
    expect(load("hi")["result.dismiss"]).not.toContain("?????");
  });
});

```

186. user (2026-06-21T19:42:48.273Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/src/i18n/catalog-invariants.test.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

187. assistant (2026-06-21T19:42:53.449Z)

Now improve `SelectionFooter.test.tsx` ? add positive localized-render assertions (exercising the kn/hi ICU plural path) and tighten the en plural check, keeping the mutation-proven negatives:

188. user (2026-06-21T19:42:56.989Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/src/components/SelectionFooter.test.tsx has

been updated successfully. (file state is current in your context ? no need to Read it back)

189. user (2026-06-21T19:43:11.901Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/src/components/SelectionFooter.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

190. assistant (2026-06-21T19:43:16.048Z)

Now add `QueryBar.test.tsx` (the cheapest real coverage win ? its handlers are untested at funcs 40%):

191. assistant (2026-06-21T19:43:26.580Z)

```
// @vitest-environment jsdom
//
// Coverage for QueryBar's interaction wiring ? the submit handler, the example-chip apply path,
the
// live parsed-pill, and the "unreadable" branch were untested (funcs 40%). The offline parser
// (lib/query) is unit-tested separately; here we prove the component glues input ? parse ?
onApply
// honestly, and that the English example chips are hidden for a non-English locale (A9).
import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
import { render, screen, cleanup, fireEvent } from "@testing-library/react";
import i18n from "../i18n";
import { QueryBar } from "../QueryBar";

beforeEach(async () => {
  window.localStorage.clear();
  await i18n.changeLanguage("en");
});
afterEach(cleanup);

describe("QueryBar", () => {
  const input = () => screen.getByLabelText(/filter rallies/i);
  const applyBtn = () => screen.getByRole("button", { name: "Apply" }) as HTMLButtonElement;

  it("a recognized query shows a parsed pill and submitting applies it as 'typed'", () => {
    const onApply = vi.fn();
    render(<QueryBar onApply={onApply} />);
    fireEvent.change(input(), { target: { value: "over 10s" } });
    expect(screen.getByTestId("query-pill").textContent).toMatch(/length > 10s/); // live
interpretation
    expect(applyBtn().disabled).toBe(false);
    fireEvent.click(applyBtn());
    expect(onApply).toHaveBeenCalledTimes(1);
    expect(onApply.mock.calls[0][1]).toBe("typed"); // source tag
  });

  it("tapping an example chip fills + applies the query as 'example'", () => {
    const onApply = vi.fn();
    render(<QueryBar onApply={onApply} />);
    fireEvent.click(screen.getByRole("button", { name: "over 10s" }));
    expect(onApply).toHaveBeenCalledTimes(1);
    expect(onApply.mock.calls[0][1]).toBe("example");
    expect((input() as HTMLInputElement).value).toBe("over 10s"); // the box is filled with the
chip text
  });

  it("an unreadable query shows the 'couldn't read that' hint and keeps Apply disabled", () => {
    const onApply = vi.fn();
    render(<QueryBar onApply={onApply} />);
    fireEvent.change(input(), { target: { value: "10s" } }); // a bare duration with no
comparator ? not recognized
```

```

    expect(screen.getByText(/Couldn't read that/i)).toBeTruthy();
    expect(applyBtn().disabled).toBe(true);
    fireEvent.submit(applyBtn().closest("form")!);
    expect(onApply).not.toHaveBeenCalled(); // submit is a no-op while unrecognized
  });

  it("hides the English-only example chips for a non-English locale (kn)", async () => {
    await i18n.changeLanguage("kn");
    render(<QueryBar onApply={vi.fn()} />);
    expect(screen.queryByRole("button", { name: "over 10s" })).toBeNull();
    await i18n.changeLanguage("en");
  });
});

```

192. user (2026-06-21T19:43:26.597Z)

File created successfully at: C:/Users/avidu/Projects/khelsutra-cov/web/src/components/QueryBar.test.tsx (file state is current in your context ? no need to Read it back)

193. assistant (2026-06-21T19:43:32.589Z)

Now run typecheck + coverage to confirm all the upgraded/new tests pass and see the new numbers (functions should rise with the QueryBar test):

194. user (2026-06-21T19:43:46.556Z)

=== TYPECHECK ===

```

> khelsutra-web@0.0.0 typecheck
> tsc --noEmit

```

exit: 0

=== TEST:COVERAGE ===

```

> khelsutra-web@0.0.0 test:coverage

```

```

> vitest run --coverage

```

```

  ? src/i18n/catalog-invariants.test.ts (24 tests)

```

```

30ms

```

```

  ? src/components/SelectionFooter.test.tsx (9 tests)

```

```

639ms

```

```

  ? SelectionFooter ? floor warning renders the audit-clean copy (en) >

```

```

allDropped + a single selection ? floorOne, passive phrasing (no 'engine'), Build blocked

```

```

414ms

```

```

Test Files  23 passed (23)

```

```

Tests       214 passed (214)

```

```

All files   | 93.34 | 88.12 | 85.88 | 93.34 |

```

```

QueryBar.tsx | 100 | 88.88 | 100 | 100 | 30

```

```

Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

```

195. assistant (2026-06-21T19:44:15.960Z)

All green: **214 tests pass, coverage up to 93.34% stmts / 88.12% branch / 85.88% funcs / 93.34% lines** (QueryBar now 100% funcs; functions metric jumped 82.35?85.88). Typecheck clean. Now I'll ratchet the functions floor up (per the review) and refresh the config comment to current numbers:

196. user (2026-06-21T19:44:19.712Z)

The file C:/Users/avidu/Projects/khelsutra-cov/web/vite.config.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

197. assistant (2026-06-21T19:44:28.865Z)

Now let me confirm the **previously-undetected** gaps are actually closed ? mutation-test the

new class-(d) guard and the self-maintaining brand rule (these were green before my fix):

198. user (2026-06-21T19:44:45.928Z)

GAP-CLOSE 1: revert hi grid.shotsConfidence???????? & grid.shots???????????? (class d)

Tests 3 failed | 21 passed

(24)

reverted

GAP-CLOSE 2: spaced-transliteration brand '??? ?????' into hi ingest.hint (evaded old denylist)

Tests 1 failed | 23 passed

(24)

reverted

GAP-CLOSE 3: reintroduce engine on errors.networkError (kn) as the Latin word 'engine'

Tests 1 failed | 23 passed

(24)

reverted

POST cleanliness:

Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

199. assistant (2026-06-21T19:45:05.283Z)

Every previously-undetected gap now fails a test (class-d: 3 failures; spaced-transliteration brand: 1; Latin "engine": 1) ? the regression net is comprehensive, and `locales` reverted clean. Final full verification (typecheck + coverage with the tightened floor + build):

200. user (2026-06-21T19:45:22.373Z)

=== TYPECHECK ===

> khelsutra-web@0.0.0 typecheck

> tsc --noEmit

tc-exit:0

=== TEST:COVERAGE (floor 92/87/84/92) ===

Test Files 23 passed (23)

Tests 214 passed (214)

All files | 93.34 | 88.12 | 85.88 | 93.34 |

cov-exit:0

=== BUILD ===

dist/assets/index-Dt-eK0X5.css

24.85 kB ? gzip: 5.43 kB

dist/assets/index-CICa-Suj.js

409.85 kB ? gzip: 124.11 kB

? built in 1.24s

Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

201. assistant (2026-06-21T19:45:31.106Z)

All green: typecheck clean, **214 tests pass, coverage 93.34/88.12/85.88/93.34 ? above the floor (92/87/84/92), gate passed (exit 0)**, build succeeds. Now commit. First re-verify branch safety (fetch + base check) and stage explicit paths:

202. user (2026-06-21T19:45:35.175Z)

=== branch ===

chore/web-coverage-floor

=== base check ===

origin/master MOVED ? rebase needed

=== origin/master HEAD ===

8328721 docs(tracker): refresh the \$7 milestone prose to merged reality (A6/A7/#77 landed) (#79)

=== status ===

M .github/workflows/test.yml

M web/package-lock.json

M web/package.json

```
M web/vite.config.ts
?? web/src/components/QueryBar.test.tsx
?? web/src/components/SelectionFooter.test.tsx
?? web/src/i18n/catalog-invariants.test.ts
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

203. assistant (2026-06-21T19:45:42.298Z)

Master moved again (docs PR #79). Let me confirm it doesn't touch my files, then commit and rebase:

204. user (2026-06-21T19:45:44.547Z)

```
=== files #79 (and anything since my base) changed ===
deploy/DEPLOY_ALPHA.md
docs/FRIENDLY_AND_MULTILINGUAL.md
=== overlap with my changeset? ===
NO overlap with web/ or .github/ ? safe rebase
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

205. assistant (2026-06-21T19:45:49.168Z)

No overlap (only docs changed). Commit my changeset, then rebase onto `origin/master`:

206. user (2026-06-21T19:46:06.989Z)

```
1119fe5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit
=== rebase onto origin/master ===
Rebasing (1/1)
```

Successfully rebased and updated refs/heads/chore/web-coverage-floor.
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra

207. assistant (2026-06-21T19:46:15.691Z)

Rebase clean. Let me verify the final state and re-run coverage once more post-rebase before pushing:

208. user (2026-06-21T19:46:26.224Z)

```
=== only my commit atop origin/master? ===
ba5e14a test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit
=== changed files ===
.github/workflows/test.yml
web/package-lock.json
web/package.json
web/src/components/QueryBar.test.tsx
web/src/components/SelectionFooter.test.tsx
web/src/i18n/catalog-invariants.test.ts
web/vite.config.ts
=== re-run coverage post-rebase ===
    Tests  214 passed (214)
All files      |  93.34 |   88.12 |   85.88 |   93.34 |
exit:0
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

209. assistant (2026-06-21T19:46:33.790Z)

Clean ? one commit, 7 files, 214 tests green, gate passes. Push and open the PR (with an honest coverage-split note per the review):

210. user (2026-06-21T19:46:36.825Z)

```
remote:
branch 'chore/web-coverage-floor' set up to track 'origin/chore/web-coverage-floor'.
To https://github.com/avidullu/khelsutra.git
* [new branch]      chore/web-coverage-floor -> chore/web-coverage-floor
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

211. user (2026-06-21T19:47:01.587Z)

```
https://github.com/avidullu/khelsutra/pull/80
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

212. assistant (2026-06-21T19:47:08.500Z)

PR opened: [khelsutra#80](https://github.com/avidullu/khelsutra/pull/80). Let me watch CI (the web job now runs the coverage gate) to completion:

213. user (2026-06-21T19:47:52.213Z)

```
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/6420b247-83bf-46ad-ae31-b9665c9bf309
Refreshing checks status every 15 seconds. Press Ctrl+C to quit.
```

```
site pass 16s
https://github.com/avidullu/khelsutra/actions/runs/27915497718/job/82600011439
web pending 0
https://github.com/avidullu/khelsutra/actions/runs/27915497718/job/82600011440
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/6420b247-83bf-46ad-ae31-b9665c9bf309
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/6420b247-83bf-46ad-ae31-b9665c9bf309
site pass 16s
https://github.com/avidullu/khelsutra/actions/runs/27915497718/job/82600011439
web pass 43s
https://github.com/avidullu/khelsutra/actions/runs/27915497718/job/82600011440
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/6420b247-83bf-46ad-ae31-b9665c9bf309
site pass 16s
https://github.com/avidullu/khelsutra/actions/runs/27915497718/job/82600011439
web pass 43s
https://github.com/avidullu/khelsutra/actions/runs/27915497718/job/82600011440
=== mergeable state ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE"}
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

214. assistant (2026-06-21T19:47:57.726Z)

CI fully green ? the `web` job ran `npm run test:coverage` and passed (the coverage gate works on CI), `mergeStateStatus: CLEAN`. Merging (squash):

215. user (2026-06-21T19:48:04.121Z)

```
=== PR state ===
{"mergeCommit":"64076a572498405c4b997a7c1df550c0e736a65b","mergedAt":"2026-06-
21T19:48:02Z","state":"MERGED"}
Shell cwd was reset to C:\Users\avidu\Projects\khelsutra
```

216. assistant (2026-06-21T19:48:11.692Z)

PR #80 **MERGED** (`64076a5`). Cleanup ? delete remote branch, remove worktree, update local master:

217. user (2026-06-21T19:48:21.721Z)

```
=== delete remote branch ===
To https://github.com/avidullu/khelsutra.git
- [deleted]          chore/web-coverage-floor
=== remove worktree ===
removed
=== update local master ===
  create mode 100644 web/src/components/SelectionFooter.test.tsx
  create mode 100644 web/src/i18n/catalog-invariants.test.ts
=== master HEAD ===
64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit (#80)
8328721 docs(tracker): refresh the S7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
```

218. assistant (2026-06-21T19:48:30.813Z)

All clean: #80 merged (`64076a5`), branch + worktree removed, local master current. Let me update the session handoff to record this checkpoint.

219. user (2026-06-21T19:48:31.527Z)

```
1      # Session handoff ? khelsutra (Friendly & Multilingual workstream)
2
3      _Last updated: 2026-06-22, after A3/A6/A7 + the kn/hi SPA-console native-review pass
(#77) merged._
4
5      ## You are here
6      - The marketing `site/` is now trilingual end-to-end and non-tech-friendly:
7      - B3 ([#70](https://github.com/avidullu/khelsutra/pull/70)): shared
`t()/`data-i18n` shim + en/kn/hi switcher + `server.js` negotiation; catalogs hoisted to repo-
root `/locales` (D-CATALOG-HOME).
8      - A3 ([#74](https://github.com/avidullu/khelsutra/pull/74)): recording-education
(homepage gist + capture checklist) + `site.` marketing copy translated to kn + hi; Noto
bundled on the site; shim gained `data-i18n-html`.
9      - A6 ([#75](https://github.com/avidullu/khelsutra/pull/75)): plain-language
Request-Access GPU/storage questions (de-jargoned + GPU reassurance), en, translation-ready.
10     - A7 ([#76](https://github.com/avidullu/khelsutra/pull/76)): accessible camera-
placement SVG diagram on the capture page, localized en/kn/hi legend.
11     - The SPA (`web/`) was already trilingual+ (en/kn/hi + es/da/id). All merged on green
CI.
12     - kn/hi translations are AI-drafted + adversarially verified ? native review still
pending before a wide public launch.
13
14     ## Next steps / open threads (tracker S7 of docs/FRIENDLY_AND_MULTILINGUAL.md)
15     - B5/B6 SPA-console native-review pass ?? ? DONE
([#77](https://github.com/avidullu/khelsutra/pull/77), squash `9f1ec9c`): fixed the adversarial-
audit kn/hi console bugs ? brand now stays Latin "KhelSutra" (dropped ????????/?????????),
"engine"/"server"/"bucket" jargon removed where en avoids it (kept where en uses it, e.g.
`ingest.uriAriaLabel`), `min_shot_count`/`<code>` token gone from `footer.floorSomeRest`, Hindi
de-Sanskritized, kn `bulk.invert`/`result.dismiss` fixed. All placeholders + `<b>/<i>/<x>`
preserved; web 177 + site 69 green; updated the kn-networkError test pin. Still AI-draft ?
native review pending (below).
16     - A2 ? localized WhatsApp share for the reel (SPA + site). Not yet started.
17     - Access form kn/hi ? A6 left the Request-Access form English on all locales (a
small future increment; `site.join.` is en-seeded).
18     - Native-speaker review of ALL AI-drafted kn/hi catalogs (site + SPA) before public
launch; owner font-glyph eyeball on a real device.
19
20     ## Ramp-up kit
```

220. user (2026-06-21T19:48:37.495Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

221. user (2026-06-21T19:48:46.857Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

222. assistant (2026-06-21T19:49:08.546Z)

Task complete. Here's the final state.

Summary

The `#77` catalog fix was already merged last turn, so "ensure coverage is improved and not regressed and merge" meant delivering the coverage discipline as a proper follow-up ? which surfaced a key fact: **`web` had no coverage measurement at all** (no provider, no thresholds, CI never gated it). You can't "not regress" what isn't measured, so I stood it up, improved it, and locked it in.

[khelsutra#80](https://github.com/avidullu/khelsutra/pull/80) ? squash-merged to master (`64076a5`) on green CI (web coverage gate ?, site ?, Cloudflare ?). Branch + worktree cleaned up; master fast-forwarded.

What landed

- **Coverage tooling + CI floor:** `@vitest/coverage-v8` + a `test:coverage` script + v8 thresholds in `vite.config.ts` (stmts/lines 92, branches 87, funcs 84). CI's web job now runs `npm run test:coverage`, so **a merge can no longer silently drop coverage**.
- **+37 tests (177 ? 214):** `SelectionFooter.test.tsx` (the floor-warning block **66% ? 100%**, doubling as a #77 render guard), `QueryBar.test.tsx` (handlers **40% ? 100% funcs**), and `catalog-invariants.test.ts` (a data-level lock for every #77 bug class).
- **Coverage: 90.44% ? 93.34% lines** (funcs 80.45 ? 85.88), gate green on CI.

Rigor

- **Adversarially reviewed** via a multi-agent workflow before merge ? it caught (and I reproduced) that my **first** draft of the invariants left the Hindi over-Sanskritization class and 7 of 10 brand keys **unguarded** (the bug could revert with tests staying green). I rewrote the guards to be self-maintaining (derived from en) and **mutation-tested** them: reintroducing each #77 bug ? engine jargon (Latin **or** transliterated), the `<code>` token, brand transliteration (incl. a spaced `??? ?????` variant that evaded the original denylist), the Hindi stiff terms ? now fails a test.
- **Honest coverage accounting** (in the PR body): the 90.44?93.34 gain is three distinct moves ? excluding non-logic files (`main.tsx`, `api/types.ts`) reframes the baseline to ~91.36; the `SelectionFooter`/`QueryBar` tests add the **real** source-line coverage to 93.34; `catalog-invariants.test.ts` adds **zero** source-line coverage (it's a content-regression guard, not part of the ratchet).

Notes for next time

- Per the concurrency convention, I worked in an isolated worktree and rebased onto `origin/master` twice (it churned: #78/#79 landed mid-task).
- Remaining coverage gap is `App.tsx` (~75%, the engine-orchestrating shell) ? a good future ratchet, noted in the handoff.
- Both PRs (#77 catalog fix, #80 coverage) are merged; nothing is left open.